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# Efficient Algorithms for Contextual Apple Tasting with Log-Loss

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## Abstract

This paper introduces two novel algorithmic approaches designed for the Contextual Apple Tasting problem, where the learner faces an asymmetric feedback structure by observing binary labels only upon an ‘Accept’ action. To address the inherent decision bias and exploration challenges, we propose two distinct but complementary strategies. First, we introduce LogCBPSide-AT, an algorithm leveraging Confidence Bounds for Partial monitoring (CBP) to explicitly quantify predictive uncertainty and effectively balance the exploration-exploitation trade-off. Second, we present LogCB-AT, an approach that reduces the apple tasting problem to an online regression oracle. This reduction-based strategy offers a computationally efficient and scalable alternative that fundamentally bypasses the complex, often intractable confidence bound constructions required by traditional methods. Theoretically, we prove that both algorithms achieve sublinear regret bounds for losses associated with the binary labels, guaranteeing robust performance even under fundamentally restricted feedback. The practical utility of our methods is empirically validated through adaptive Large Language Model (LLM) cascading, where they effectively optimize the trade-off between inference cost and response accuracy.

## 1 Introduction

In the field of sequential decision-making, the standard contextual bandit framework serves as a foundational model for learning under uncertainty. Within this setup, a learner observes a context and selects an action, receiving feedback only for the chosen arm. Achieving high performance in such tasks depends on effectively managing the balance between exploration and exploitation.

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The agent must decide whether to exploit its greedy action to maximize immediate rewards or explore alternative actions to gather information that enables more accurate future decision-making. While this standard framework has proven effective across a wide range of practical problems [Li et al., 2010, Tewari and Murphy, 2017, Durand et al., 2018], more challenging settings arise when the information structure is fundamentally restricted. This challenge is especially prominent in environments characterized by partial monitoring, where feedback may be entirely absent for certain choices. A particularly notable subclass of this setting is the Apple Tasting problem, originally introduced by Helmbold et al. [2000]. In this framework, the learner faces an asymmetric information structure where the true label of an instance (e.g., whether the apple is rotten or not) is revealed only if the learner decides to 'Accept' (or 'Taste') it. If the learner chooses to remain 'Reject' (or 'Pass'), no feedback is provided, leaving the true status of the instance unknown. The learner's goal is to minimize the accumulated loss induced by the interaction between its actions (Taste vs. Pass) and the binary labels (rotten vs non-rotten), e.g., reducing the number of untasted rotten apples and tasted non-rotten apples. This objective faces a fundamental challenge where errors in rejection (false negatives) are never explicitly corrected, potentially leading to a persistent bias where the model remains stuck in a suboptimal state due to the continuous absence of corrective feedback.

In this paper, we study the contextual apple tasting problem, where action selection is informed by contextual information. The decision to Accept reveals the binary label at a specific cost, while the Reject arm does not provide any feedback, and potentially results in large loss according to the value of  $y_t$ . This setup is motivated by modern AI application scenarios, for which we present two representative examples below:

**LLM cascading** The system initially processes a query through a Small Language Model (SLM) and must determine whether to cascade to a Large Language Model (LLM) for a more refined result. While the SLM provides an efficient baseline response, complex queries may require the LLM's superior reasoning at the cost of significantly higher resources and latency. Since the true performance gain is only observable upon invoking the LLM, the system operates under partial feedback: staying with the SLM prevents learning whether the LLM would have produced a better answer. Optimizing this trade-off is critical to avoid either unnecessary computational cost or degraded user satisfaction from suboptimal SLM outputs. In this example, invoking LLM and staying with SLM correspond to 'Accept' and 'Reject' actions respectively, and whether a performance gain was obtained corresponds to binary label. The sequence of queries serves as contextual information.

**Digital Healthcare** Wearable devices monitor bio-signals (contextual information) to determine if a patient needs a clinical visit. This is a quintessential contextual apple tasting problem because the actual presence of a disease (binary label) can usually only be confirmed if the device has notified the patient to see a doctor ('Accept'). If the device remains silent ('Reject'), the true medical status remains unobserved. In this context, failing to notify a patient who is actually ill results in an exceptionally large loss. Conversely, notifying a healthy patient leads to significant patient inconvenience and unnecessary medical expenses.

The asymmetric feedback structure necessitates a sophisticated exploration strategy to prevent the model from becoming permanently biased due to missing data. In this paper, we propose two distinct but complementary algorithmic approaches that efficiently tackle this challenge: LogCBPSide-AT, an confidence bound algorithm based on a logistic model assumption on the label, and LogCB-AT, a reduction-to-regression-oracle approach that accommodates any model class that can be learned under the log-loss objective. The two algorithms cater to different modeling assumptions and theoretical guarantees: the former employs a refined exploration-exploitations strategy within a more restrictive model class with possibly logarithmic regret, while the latter accommodates a broader class of models with  $O(\sqrt{T})$  regret, where  $T$  is the time horizon.

First, we develop LogCBPSide-AT, an algorithm leveraging Confidence Bounds for Partial monitoring (CBP) to explicitly quantify predictive uncertainty under the logistic model. While this approach provides a principled and theoretically rigorous mechanism for balancing exploration and exploitation under partial feedback [Bartók and Szepesvári, 2012, Heuillet et al., 2024], constructing explicit confidence bounds can become computationally demanding when scaling to complex function classes. To provide a scalable alternative that accommodates a broader function class, including the logistic model, we subsequently introduce LogCB-AT, a reduction-based approach that adapts

the SquareCB algorithm of Foster and Rakhlin [2020] to the binary, partial feedback setting. This approach fundamentally bypasses the challenge of explicit bound construction by reducing the bandit learning task to repeated calls to an online regression oracle with a log-loss objective. Consequently, it eliminates the need for complex exploration terms and allows for straightforward integration of black-box or complex predictive models. Importantly, both algorithms, LogCBPSide-AT and LogCB-AT, adapt their respective base algorithms—CBPSide and SquareCB—to the one-sided feedback structure, by exploiting the structural property of the Apple Tasting problem: the latent true label  $y_t$  is a shared ground truth independent of the learner’s action.

## 2 Problem Setting and Preliminaries

We consider a contextual binary decision problem over  $T$  rounds. At each round  $t \in [T] := \{1, \dots, T\}$ , the learner observes a context vector  $x_t \in \mathcal{X} \subset \mathbb{R}^d$  and must choose an action  $a_t \in \mathcal{A} := \{0, 1\}$ , where 0 and 1 denote ‘Reject’ and ‘Accept’, respectively. After the action is chosen, a latent binary label  $y_t \in \{0, 1\}$  is realized according to an unknown conditional distribution. The defining feature of apple tasting is its one-sided feedback structure. Specifically, the true label  $y_t$  is revealed only when the learner chooses the revealing action ( $a_t = 1$ ); if the learner chooses the Reject action ( $a_t = 0$ ), no feedback is observed.

Our basic assumption is realizability: the learner has access to a function class  $\mathcal{F} \subseteq (\mathcal{X} \rightarrow [0, 1])$  that contains the true conditional expectation of the binary label.

**Assumption 1** (Realizability). *There exists a function  $f^* \in \mathcal{F}$  such that, for all  $x \in \mathcal{X}$ ,*

$$f^*(x) = \mathbb{E}[y \mid x].$$

As for LogCBPSide-AT, we further assume (in Section 4) that  $\mathcal{F}$  is the class of logistic functions, whereas for LogCB-AT, we impose no additional restriction beyond the existence of a regression oracle that can efficiently learn  $f^*$  under a log-loss objective (Assumption 2 in Section 5). We denote  $f^*(x_t) = \mathbb{E}[y_t \mid x_t]$  by  $y_t^*$ , the conditional mean at round  $t$ . The learner maintains an estimate  $\hat{f}_{t-1}$  of  $f^*$ , and its prediction at round  $t$  is  $\hat{y}_t := \hat{f}_{t-1}(x_t)$ . The learner is evaluated through the asymmetric loss matrix

|         |             |             |
|---------|-------------|-------------|
|         | $y = 0$     | $y = 1$     |
| $a = 0$ | 0           | $\ell_{01}$ |
| $a = 1$ | $\ell_{10}$ | $\ell_{11}$ |

where  $\ell_{01}, \ell_{10}, \ell_{11} \in [0, 1]$ . Here,  $\ell_{01}$  is the loss of rejecting an item with label 1,  $\ell_{10}$  is the loss of accepting an item with label 0, and  $\ell_{11}$  is the cost incurred when accepting an item with label 1. Throughout the paper, we assume the following cost conditions

$$\ell_{01} > \ell_{11} \geq 0 \quad \text{and} \quad \ell_{10} > 0.$$

The strict inequality  $\ell_{01} > \ell_{11}$  is necessary to prevent the Reject action ( $a = 0$ ) from trivially always being the optimal choice. This aligns naturally with practical scenarios like LLM cascading: the performance penalty of missing a better answer ( $\ell_{01}$ ) inherently outweighs the cost of invoking the LLM ( $\ell_{11}$ ). Additionally,  $\ell_{11} \geq 0$  simply reflects that any invocation incurs a basic cost, while  $\ell_{10} > 0$  ensures that an unnecessary call strictly incurs a computational penalty.

For any scalar  $z \in [0, 1]$ , we define the loss function as  $\psi_a(z) := \ell_{01}z(1-a) + \{\ell_{10} + (\ell_{11} - \ell_{10})z\}a$ . For label  $y$ ,  $\psi_0(y)$  and  $\psi_1(y)$  represent the losses of rejection and acceptance, respectively. Also, since  $\psi_a(\cdot)$  is linear in  $z$  for each  $a \in \mathcal{A}$ , we have,

$$\mathbb{E}[\psi_a(y_t) \mid x_t] = \psi_a(f^*(x_t)) = \psi_a(y_t^*).$$

The optimal action at round  $t$  is defined as

$$a_t^* \in \arg \min_{a \in \mathcal{A}} \psi_a(y_t^*),$$

i.e., the action with the smaller conditional expected loss under the true model. The instantaneous regret is then defined by

$$r_t := \psi_{a_t}(y_t) - \psi_{a_t^*}(y_t),$$

and the cumulative regret is  $R(T) := \sum_{t=1}^T r_t$ . In Sections 4 and 5, we present two novel algorithms, LogCBPSide-AT and LogCB-AT, respectively, both of which achieve  $O(\sqrt{T})$  upper bounds on  $R(T)$  with high probability under their respective assumptions. As for LogCBPSide-AT, we also present instance-dependent regret bound with polylogarithmic order in  $T$ .

### 3 Related work

Partial monitoring (PM) studies sequential decision problems where the learner does not observe the loss directly, but only a feedback signal determined by the chosen action and the hidden label. In stochastic finite PM, games are classically categorized into trivial, easy, hard, and hopeless classes according to their minimax regret rates [Bartók and Szepesvári, 2012, Bartók et al., 2014]. Apple tasting is one of the canonical PM examples, and is known to be a two-action, two-outcome easy game, for which the minimax regret scales on the order of  $\sqrt{T}$  [Heuillet et al., 2024].

Helmbold et al. [2000] was the first to formalize the apple tasting model, demonstrating that an  $O(\sqrt{T})$  mistake bound is achievable through randomized algorithms in the realizable setting, where the target hypothesis is assumed to reside within the hypothesis class. This foundational work established the essential framework for transforming standard online learning models into the apple tasting paradigm. Subsequently, Raman et al. [2024] extended this research from a combinatorial perspective, providing a comprehensive analysis across both realizable and agnostic settings. By introducing a new combinatorial parameter known as ‘Effective Width,’ they established a trichotomy for mistake bounds in the realizable setting, showing that the expected number of mistakes must fall into one of three categories:  $\Theta(1)$ ,  $\Theta(\sqrt{T})$ , or  $\Theta(T)$ . Notably, they resolved a long-standing open question by characterizing agnostic learnability via the Littlestone dimension. In contrast to previous literature that primarily focused on randomized algorithms, Chase and Mehalé [2024] explored the feasibility of deterministic algorithms within the realizable setting. They proved that any hypothesis class learnable under Apple tasting feedback is necessarily learnable by a deterministic algorithm. By providing an optimal mistake bound of  $O(\sqrt{L(\mathcal{H})T \log T})$ , where  $L(\mathcal{H})$  denotes the Littlestone dimension of the hypothesis class  $\mathcal{H}$ , this study completed the theoretical foundation for deterministic Apple tasting.

While the aforementioned literature established the theoretical foundation of apple tasting in non-contextual settings, subsequent research expanded this framework to contextual bandits, where side information is available for decision-making. Bartók and Szepesvári [2012] extended the confidence bound based method of Bartók et al. [2012] for partial monitoring to the contextual setting, introducing CBPSide. More recently, Heuillet et al. [2024] proposed RandCBP and RandCBPSide by incorporating randomization into these frameworks to improve exploration efficiency. Bayesian strategies have also been adapted to better balance the exploration-exploitation trade-off in partial monitoring. Tsuchiya et al. [2020] established theoretical regret bounds for Thompson Sampling (TS), while Kirschner et al. [2020] proposed Information-Directed Sampling (IDS) to minimize the information ratio. Specifically, Grant and Leslie [2021] revisited the apple tasting problem, demonstrating the empirical superiority of these Bayesian mechanisms in managing restricted feedback. In addition, Harris et al. [2023] proposed an EXP-based algorithm to address apple tasting in adversarial scenarios. While LogCBPSide-AT is adapted from CBPSide, it is simplified and specialized for the logistic contextual apple tasting setting. Beyond this, we establish an instance-dependent regret bound for LogCBPSide-AT, a guarantee that, to our knowledge, has not been previously derived for CBPSide. We note that the partial monitoring literature predominantly focuses on minimax bounds; among the few works that establish instance-dependent guarantees, the MED-based approach of Komiya et al. does not prove a minimax bound, whereas LogCBPSide-AT achieves both.

Recent contextual bandit research has increasingly explored reductions to online regression oracles, beginning with SquareCB [Foster and Rakhlin, 2020]. Within this line of work, Zhang et al. [2023] proposed SquareCB.G, adapting SquareCB [Foster and Rakhlin, 2020] to the partial monitoring setting, including Apple Tasting. However, the SquareCB.G framework is restricted to the use of a square-loss oracle, which may not be well-suited for binary labels, as in the original Apple Tasting problem and the aforementioned modern applications in LLMs and Digital Healthcare. In this work, we adapt the SquareCB algorithm to the binary feedback setting using a log-loss oracle. The derivation of the action-selection policy fundamentally differs from that of SquareCB.G., resulting in a fully randomized algorithm, as opposed to the partially deterministic policy used in SquareCB.G.

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**Algorithm 1** LogCBPSide-AT

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**Input:**  $\lambda, \alpha, \{\delta_t\}$   
Observe the context  $x_1$ , Play action  $a_1 = 1$  (Accept) and Observe  $y_1$   
 $N_1 = 1$  ( $N_t$  is the number of times the action Accept ( $a_t = 1$ ) is chosen up to time  $t$ )  
 $V_1 = \lambda I + x_1 x_1^T$   
Compute  $\hat{\theta}_1$  such that  $(y_1 - \sigma(x_1^T \hat{\theta}_1)) x_1 = 0$   
**for**  $t > 1$  **do**  
  Observe the context  $x_t$ , Calculate  $\hat{y}_t = \sigma(x_t^T \hat{\theta}_{t-1})$   
  Let  $\beta_{t-1}^{x_t}(\delta_t) = \frac{1}{2c_\sigma} \|x_t\|_{V_{t-1}^{-1}} \sqrt{\left(3 + 2 \log\left(1 + \frac{2}{\lambda}\right)\right) \cdot 2d \log(N_{t-1}) \cdot \log\left(\frac{d}{\delta_t}\right)}$   
  **if**  $|\hat{y}_t - \alpha| < \beta_{t-1}^{x_t}(\delta_t)$  **then**  
    Exploration phase  
    Play action  $a_t = 1$  (Accept) and observe  $y_t$   
  **else**  
    Exploitation phase  
    **if**  $\hat{y}_t \leq \alpha$  **then**  
      Play action  $a_t = 0$  (Reject)  
    **else**  
      Play action  $a_t = 1$  (Accept) and observe  $y_t$   
    **end if**  
  **end if**  
  Update  $V_t = V_{t-1} + x_t x_t^T \cdot \mathbb{1}\{a_t = 1\}$   
  Compute  $\hat{\theta}_t$  such that  $\sum_{s=1}^t \mathbb{1}\{a_s = 1\} (y_s - \sigma(x_s^T \hat{\theta}_t)) x_s = 0$   
   $N_t = N_{t-1} + \mathbb{1}\{a_t = 1\}$   
**end for**

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#### 4 Confidence bound-based algorithm: LogCBPSide-AT

In this section, we present LogCBPSide-AT (Algorithm 1), a deterministic algorithm for the logistic contextual apple tasting problem, adapted from CBPSide Bartók and Szepesvári [2012]. We analyze logistic function class,  $f^*(x) = \mathbb{E}[y | x] = \sigma(x^T \theta^*)$ . The algorithm maintains an estimate of  $\theta^*$  (maximum quasi-likelihood estimator) alongside a confidence interval of width  $\beta_{t-1}^{x_t}(\delta_t)$  (defined in Algorithm 1) for the unknown latent label  $y_t^*$  given context  $x_t$  and selects actions based on the relationship between this confidence interval and a decision threshold.

We define the flip point  $\alpha \in (0, 1)$  as the value at which the expected losses of actions Accept and Reject are equal:

$$\psi_0(\alpha) = \psi_1(\alpha) \implies \alpha = \ell_{10} / (\ell_{10} + \ell_{01} - \ell_{11})$$

For any context  $x_t$ , action  $a = 0$  (Reject) is the optimal action whenever  $y_t^* < \alpha$ , and action  $a = 1$  (Accept) is optimal otherwise. Accordingly, when the confidence interval lies entirely to one side of  $\alpha$ , LogCBPSide-AT acts greedily by exploiting its current estimate  $\hat{y}_t$  to select the optimal action. When the confidence interval overlaps with  $\alpha$ , the algorithm resolves this uncertainty by selecting Accept, thereby obtaining feedback to refine future estimates and confidence bounds. We present the full procedure for LogCBPSide-AT in Algorithm 1.

The theoretical analysis of LogCBPSide-AT is deferred to Appendix C. We state the main regret guarantee below. LogCBPSide-AT achieves the minimax-optimal regret bound of order  $\sqrt{T}$ , consistent with the known lower bounds for easy partial monitoring games with side information.

**Theorem 1** (LogCBPSide-AT minimax regret bound). *Assume  $\sup_{x \in \mathcal{X}} \|x\|_2 \leq 1$  and  $\sup_{x \in \mathcal{X}} |x^T \theta^*| \leq C_{\max}$ . For  $\delta_t = \frac{1}{t}$ , the expected cumulative regret of the LogCBPSide-AT*

algorithm is bounded as:

$$\mathbb{E} [Reg_T] \leq O \left( \left(1 + e^{C_{\max}}\right)^2 e^{-C_{\max}} d \sqrt{T} \log^{\frac{3}{2}}(T) \right).$$

(Ignoring logarithmic factors in  $d$ )

Furthermore, an important advantage of confidence-based algorithms in online learning with limited feedback is that they often enjoy instance dependent regret bound of order  $\text{polylog}(T)$ . We prove that LogCBPSide-AT also enjoys a  $\log^2 T$  regret bound (see Theorem 2), a guarantee that, to our knowledge, has not been previously derived for CBPSide. Proof of Theorem 2 is deferred to Appendix C.2.

**Theorem 2** (LogCBPSide-AT instance dependent regret bound). *Assume  $\sup_{x \in \mathcal{X}} \|x\|_2 \leq 1$ ,  $\sup_{x \in \mathcal{X}} |x^T \theta^*| \leq C_{\max}$ , and the existence of  $\Delta > 0$  such that  $\Delta \leq \min_{x \in \mathcal{X}} |\alpha - \sigma(x^T \theta^*)|$ . For  $\delta_t = \frac{1}{t}$ , the expected cumulative regret of the LogCBPSide-AT algorithm is bounded as:*

$$\mathbb{E} [Reg_T] \leq O \left( \frac{1}{\Delta} d^2 \log^2 T \right)$$

(ignoring doubly logarithmic factors and  $\text{polylog}(\frac{1}{\Delta})$ )

## 5 SquareCB-based algorithm: LogCB-AT

We now introduce **LogCB-AT**, our adaptation of SquareCB to the contextual apple tasting setting. Our algorithm relies on an online regression oracle, denoted by  $\text{Alg}_{\text{KL}}$ , which produces a prediction  $\hat{y}_t = \hat{f}_{t-1}(x_t)$  at round  $t$ . Since the true label is revealed only when the learner takes the Accept action ( $a_t = 1$ ), the oracle is updated only on those rounds.

**Assumption 2** (Logarithmic-loss oracle guarantee). *For any (possibly adaptive) sequence of contexts, actions, and labels, the online log loss regression oracle  $\text{Alg}_{\text{KL}}$  satisfies*

$$\sum_{t=1}^T \mathbf{1}\{a_t = 1\} \ell_{\log}(\hat{y}_t, y_t) - \inf_{f \in \mathcal{F}} \sum_{t=1}^T \mathbf{1}\{a_t = 1\} \ell_{\log}(f(x_t), y_t) \leq \text{Reg}_{\text{KL}}(T_1) \leq \text{Reg}_{\text{KL}}(T),$$

where  $\hat{y}_t = \hat{f}_{t-1}(x_t)$ ,  $\ell_{\log}$  denotes the standard log-loss (binary cross-entropy),

$$\ell_{\log}(\hat{y}, y) = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})],$$

and  $T_1 := \sum_{t=1}^T \mathbf{1}\{a_t = 1\}$  denotes the number of rounds on which the action  $a_t = 1$  is taken.

Our proposed algorithm, LogCB-AT, is presented in Algorithm 2. It closely follows the structure of SquareCB. At each round  $t$ , the learner first computes the greedy action  $b_t$  and defines the action-selection policy which a learning rate  $\gamma > 0$  and an exploration parameter  $\mu$ . The learner then samples an action  $a_t$ , observes the label  $y_t$  only if  $a_t = 1$ , and updates  $\text{Alg}_{\text{KL}}$  only on those rounds. A notable difference from SquareCB is that (i) the oracle uses a log-loss objective rather than a square-loss objective, and (ii) the oracle is updated only on rounds with  $a_t = 1$ . These differences call for non-trivial regret analysis and a new formula for the optimal value of  $\gamma$  which enables an  $O(\sqrt{T})$  regret bound under mild guarantees of the oracle. We elaborate on details in the following subsections.

We analyze the regret of LogCB-AT algorithm, and propose a new formula for the learning rate  $\gamma$  which effectively minimizes the upper bound of the regret under the challenge of one-sided, binary feedback. Our analysis begins by partitioning the time steps  $[T]$  into two disjoint sets,  $\mathcal{I}_1$  and  $\mathcal{I}_0$ , according to the greedy action  $b_t$ :  $\mathcal{I}_1 := \{t \in [T] : b_t = 1\}$  and  $\mathcal{I}_0 := \{t \in [T] : b_t = 0\}$ . The regret is then analyzed separately over the rounds in  $\mathcal{I}_1$  and  $\mathcal{I}_0$ .

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**Algorithm 2** LogCB-AT

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**Require:**  $\gamma > 0, \mu > 0$ , online regression oracle for log loss  $\text{Alg}_{\text{KL}}$

```
1: for  $t = 1, \dots, T$  do
2:   Observe context  $x_t \in \mathcal{X}$ 
3:   Obtain prediction  $\hat{y}_t = \hat{f}_{t-1}(x_t)$  from  $\text{Alg}_{\text{KL}}$ 
4:   Set  $b_t \in \arg \min_{a \in \{0,1\}} \psi_a(\hat{y}_t)$ 
5:   Set
      
$$p_{t,a} = \frac{1}{\mu + \gamma(\psi_a(\hat{y}_t) - \psi_{b_t}(\hat{y}_t))} \text{ for } a \neq b_t, \quad p_{t,b_t} = 1 - \sum_{a \neq b_t} p_{t,a}$$

6:   Sample  $a_t \sim p_t$ 
7:   if  $a_t = 1$  then
8:     Observe  $y_t$  and update  $\text{Alg}_{\text{KL}}$  with  $(x_t, y_t)$ 
9:   end if
10: end for
```

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**Regret for  $t \in \mathcal{I}_1$**  For  $t \in \mathcal{I}_1$ , the regret is decomposed according to the following corollary, which originates from Foster and Rakhlin [2020].

**Corollary 1.** *Suppose Assumptions 1 hold. With probability at least  $1 - \delta$ , we have*

$$\begin{aligned} \sum_{t \in \mathcal{I}_1} r_t &\leq \sum_{t \in \mathcal{I}_1} \sum_{a \in \mathcal{A}} p_{t,a} \left[ (\psi_a(y_t^*) - \psi_{a_t^*}(y_t^*)) - \frac{\gamma}{4} (\psi_a(\hat{y}_t) - \psi_{a_t}(\hat{y}_t))^2 \right] \\ &\quad + \frac{\gamma}{4} \sum_{t \in \mathcal{I}_1} \sum_{a \in \mathcal{A}} p_{t,a} (\psi_a(\hat{y}_t) - \psi_{a_t}(\hat{y}_t))^2 + \max(\ell_{10}, \ell_{01} - \ell_{11}) \sqrt{2T \log(2\delta^{-1})}. \end{aligned}$$

The first term of the right-hand side is upper bounded by  $4T/\gamma$  under the action selection policy specified in Algorithm 2, with specific choice of  $\mu = 2$ . The challenge arises in the second term, which represents the squared loss of the prediction  $\hat{y}_t$  over all rounds in  $\mathcal{I}_1$ , whereas Assumption 2 guarantees an upper bound only on the cumulative log loss over rounds with  $a_t = 1$ . To circumvent this issue, we exploit the structural property of Apple Tasting that, at each time  $t$ , both actions share a common latent label  $y_t$ , and leverage a probability-shift argument between  $p_{t,0}$  and  $p_{t,1}$  for rounds in  $\mathcal{I}_1$ . Finally, we employ the conversion between squared loss and log-loss used in Foster and Krishnamurthy [2021] to bound the second term as follows,

$$\frac{\gamma}{4} \sum_{t \in \mathcal{I}_1} \sum_{a \in \mathcal{A}} p_{t,a} (\psi_a(\hat{y}_t) - \psi_{a_t}(\hat{y}_t))^2 \leq \frac{\gamma}{4} \Lambda_1 \left( 8\text{Reg}_{\text{KL}}(T) + 4 \log(2\delta^{-1}) \right),$$

where  $\Lambda_1 = (\ell_{10} - \ell_{11})^2 + \ell_{01}^2/(\mu - 1)$ . The complete proofs are deferred to Appendix A. We note that choosing  $\gamma = O(\sqrt{T/\text{Reg}_{\text{KL}}(T)})$  yields a bound of  $O(\sqrt{T\text{Reg}_{\text{KL}}(T)})$  for the cumulative regret over rounds in  $\mathcal{I}_1$ .

**Regret for  $t \in \mathcal{I}_0$**  A similar technique to that used for analyzing the regret over rounds in  $\mathcal{I}_1$  could also be employed to bound the regret for  $t \in \mathcal{I}_0$ . However, when  $b_t = 0$ , the probability ratio  $p_{t,0}/p_{t,1}$  introduces an additional dependence on  $\gamma$  in the second term. As a result, setting  $\gamma = O(\sqrt{T/\text{Reg}_{\text{KL}}(T)})$  leads to a regret bound of  $O(T\text{Reg}_{\text{KL}}(T))$ . The optimal choice of  $\gamma$  for rounds in  $\mathcal{I}_0$  instead turns out to be  $O\left(\{T/\text{Reg}_{\text{KL}}(T)\}^{1/3}\right)$ , which yields a suboptimal regret bound of  $O\left(\{T\text{Reg}_{\text{KL}}(T)\}^{2/3}\right)$ . To achieve  $O(\sqrt{T\text{Reg}_{\text{KL}}(T)})$  regret for rounds in  $\mathcal{I}_0$  as well, we devise a novel relationship between the regret and prediction loss that fundamentally differs from Corollary 2. Specifically, we further decompose the rounds in  $\mathcal{I}_0$  according to whether  $a_t^* = 1$  or  $a_t^* = 0$ , and derive the following inequalities for each case:

For  $t \in \mathcal{I}_0 \cap \{t \in [T] : a_t^* = 1\}$ ,

$$\mathbb{E}[r_t] \leq C_{\Delta} \mathbb{E}[I(a_t = 1)|y_t^* - \hat{y}_t|] + \gamma C_{\Delta}^2 \mathbb{E}[I(a_t = 1)(y_t^* - \hat{y}_t)^2],$$

and for  $t \in \mathcal{I}_0 \cap \{t \in [T] : a_t^* = 0\}$ ,

$$\mathbb{E}[r_t] \leq \frac{1}{\gamma} + C_\Delta \mathbb{E}[I(a_t = 1)|\hat{y}_t - y_t^*|],$$

where  $C_\Delta = |\ell_{01} - \ell_{11} + \ell_{10}|$ . We note that neither inequality involves a quadratic dependence on  $\gamma$ , and moreover, the prediction loss in the right-hand side is accumulated only over rounds with  $a_t = 1$ . Consequently, the cumulated regret over rounds in  $\mathcal{I}_0$  enjoys the same  $O(\sqrt{T \text{Reg}_{KL}(T)})$  bound with the same choice of  $\gamma$ . Detailed proof is provided in Appendix A.

Building upon the analysis above, we present the regret bound for the LogCB-AT algorithm.

**Theorem 3** (LogCB-AT Regret Bound). *Suppose Assumptions 1 and 2 hold. For the choice of  $\gamma = O(\sqrt{T/\text{Reg}_{KL}(T)})$ , the cumulative regret of the LogCB-AT algorithm is bounded by:*

$$R(T) = O\left(\sqrt{T \cdot \text{Reg}_{KL}(T)}\right)$$

*with probability at least  $1 - \delta$ .*

**Remark.** LogCB-AT guarantees strictly positive, closed-form probabilities for all actions. Under the standard cost assumption  $l_{10} \geq l_{11} \geq 0$ ,  $l_{01} > 0$ , the maximum loss gap is bounded by  $g_{\max} = \max(l_{01} - l_{11}, l_{10})$ . Consequently, the non-greedy action probability is strictly bounded within  $[\frac{1}{2+\gamma \cdot g_{\max}}, \frac{1}{2}]$ , and the greedy action probability is bounded within  $[\frac{1}{2}, 1 - \frac{1}{2+\gamma \cdot g_{\max}}]$ . This fully randomized design contrasts with partially deterministic methods such as SquareCB.G and CBPSide, which may assign zero probability to non-greedy actions. This distinction is particularly important for Off-Policy Evaluation (OPE), where estimators such as IPW require non-zero action probabilities for valid counterfactual inference [Bian and Jun, 2022, Balagopalan and Jun, 2025]. By explicitly lower bounding the action probabilities, LogCB-AT avoids unstable IPW weights and enables robust offline evaluation.

**Comparison of LogCBPSide-AT and LogCB-AT** In this work, we proposed two complementary algorithms for the contextual Apple Tasting problem with provable guarantees under respective assumptions. In particular, LogCBPSide-AT exhibits strong theoretical guarantees, with the possibility of achieving logarithmic regret under suitable assumptions in the logistic-linear setting. On the other hand, LogCB-AT does not attain logarithmic regret, but instead offers greater flexibility by accommodating richer hypothesis classes beyond the logistic-linear setting. Therefore, when a good and rich representation is available and the logistic-linear assumption is appropriate, LogCBPSide-AT may be preferred due to its stronger guarantees. In contrast, when the logistic-linear assumption is hard to validate, LogCB-AT provides a more flexible alternative. In this sense, the two approaches serve as complementary alternatives, each being advantageous under different modeling assumptions and application scenarios.

## 6 Experiments

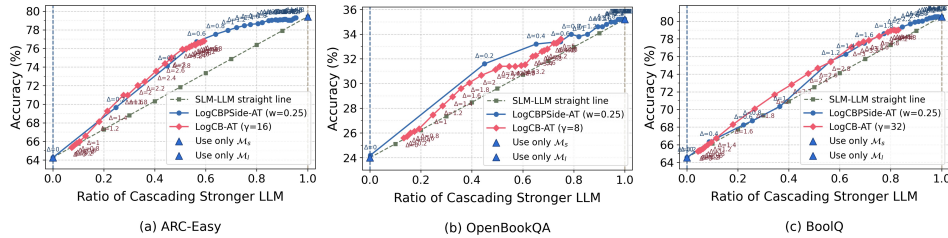


Figure 1: Accuracy vs. LLM Invocation Ratio

**Task Formulation** In this section, we empirically evaluate the performance of the proposed LogCB-AT and LogCBPSide-AT algorithms on the practical real-world application of test-time adaptive



Large Language Model (LLM) cascading, which we frame as a contextual Apple Tasting problem, namely AppleTeA: Test-Time Adaptive LLM Cascading via Logistic Apple Tasting. The goal is to dynamically route streaming queries to either a computationally efficient Small Language Model (SLM) or a highly capable Large Language Model (LLM), thereby optimizing the trade-off between inference cost and response accuracy. At each time step  $t$ , the system receives a query  $q_t$  and first generates a response using the SLM ( $\mathcal{M}_s$ ). The algorithm then makes a binary decision  $a_t \in \{1, 0\}$ : choosing  $a_t = 0$  (Reject) finalizes the SLM response without additional cost, while choosing  $a_t = 1$  (Accept) invokes the stronger LLM ( $\mathcal{M}_l$ ) at a higher computational cost. Since the ground-truth is inaccessible during test-time inference, we rely on the highly capable LLM as an oracle to provide pseudo-labels. Under the practical assumption that the stronger LLM produces correct or superior answers, a discrepancy between the SLM and LLM responses is treated as an SLM prediction failure. Thus, the pseudo-label is defined as  $y_t = 1$  if the response generated by  $\mathcal{M}_s$  differs from that of  $\mathcal{M}_l$ , and  $y_t = 0$  otherwise. Importantly, this label is observed only when  $a_t = 1$ , naturally inducing a partial-feedback apple tasting structure. We carefully construct the loss matrix to encode two competing cost components 1) the invocation cost of the strong LLM (i.e., token-level inference cost) and 2) the price of error incurred when the weak SLM is invoked in contexts where prediction failure is likely. Further details on the experiments, including the construction of contexts, datasets, and the choice of SLM/LLM models, are provided in Appendix D.

**Baselines and Evaluation Metrics.** We compare our algorithms against three baselines: (i)  $\mathcal{M}_s$ -only, (ii)  $\mathcal{M}_l$ -only, and (iii) random cascading with a fixed probability  $p_{\text{random}} \in [0, 1]$ . In the figure, the x-axis represents the proportion of total queries routed to the stronger LLM. Moving to the right indicates an increase in invocations, which directly translates to higher operational costs. The y-axis reflects the final accuracy resulting from these routing decisions.

**Experimental Results** To demonstrate the algorithmic efficiency of LogCB-AT and LogCBPSide-AT, we evaluated the trade-off between LLM invocation rate and accuracy. We fixed the internal exploration hyperparameters (learning rate  $\gamma$  for LogCB-AT and a confidence scaling factor  $w$  for LogCBPSide-AT) to the values selected via grid search, and the penalty cost of invoking LLM when  $y_t = 0$  ( $l_{10}$ ) to 1.0. We then plotted the average accuracy against the LLM invocation rate for each algorithm by varying the net penalty of SLM failure ( $l_{01} - l_{11}$ ), while keeping all other parameters fixed. The reported results were averaged over 5 independent runs.

Figures 1 show that varying this cost parameter produces concave trade-off curves. This confirms that, rather than blindly exploring, both algorithms efficiently learn the contextual mapping under partial feedback, invoking the LLM strategically only when the SLM is likely to fail. Ultimately, both algorithms effectively manage the exploration-exploitation trade-off, consistently maintaining optimal cost-efficiency across varying cost objectives.

## 7 Conclusion

In this paper, we introduced two novel algorithms, LogCBPSide-AT and LogCB-AT, designed to tackle the contextual Apple Tasting problem. Specifically, LogCBPSide-AT specializes the partial monitoring-based CBPSide framework for the Apple Tasting scenario by introducing a logistic setting. Concurrently, LogCB-AT adapts the contextual bandit-based SquareCB algorithm to operate within a log-loss setting. By leveraging these frameworks, we provided rigorous theoretical guarantees, demonstrating that our algorithms achieve sublinear regret bounds. Beyond the theoretical formulations, we empirically validated the practical utility of our approaches in a real-world application of test-time adaptive Large Language Model (LLM) cascading. Our experiments showed that dynamically routing queries between a computationally efficient Small Language Model (SLM) and a highly capable LLM via these algorithms effectively optimizes the trade-off between inference cost and response accuracy. Building upon these foundational results, our future work aims to extend this framework to achieve first-order regret bounds. By adapting the FastCB algorithm [Foster and Krishnamurthy, 2021], we plan to develop a more advanced variant that scales with the loss of the optimal policy, yielding tighter regret guarantees and further accelerating convergence in practical environments.

## **8 Acknowledgement**

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# Appendix

## A LogCB-AT Regret Analysis

In this section, we derive a high-probability regret bound for the LogCB-AT algorithm. We aim to show that with probability at least  $1 - \delta$ , the cumulative regret is bounded by  $O(\sqrt{T \cdot \text{Reg}_{\text{KL}}(T)})$ .

### A.1 Definition of Regret

The instantaneous regret at round  $t$  is defined as the difference between the loss of the action taken  $a_t$  and the loss of the optimal action  $a_t^*$ .

$$r_t = \psi_{a_t}(y_t) - \psi_{a_t^*}(y_t). \quad (1)$$

The cumulative regret  $R(T)$  over  $T$  rounds is the sum of these instantaneous regrets.

$$R(T) = \sum_{t=1}^T r_t. \quad (2)$$

**Lemma 1.** For any  $\hat{y}, y^* \in [0, 1]$ , the KL-divergence between two Bernoulli distributions with parameters  $\hat{y}$  and  $y^*$  satisfies:

$$d_{\text{KL}}(\hat{y}, y^*) \geq \frac{1}{2} \frac{(\hat{y} - y^*)^2}{\hat{y} + y^*}. \quad (3)$$

Consequently, the following pointwise inequality holds for all  $\hat{y}, y^* \in [0, 1]$ :

$$(\hat{y} - y^*)^2 \leq 4d_{\text{KL}}(\hat{y}, y^*). \quad (4)$$

*Proof.* The first inequality is a known lower bound for the KL-divergence of Bernoulli distributions (see, e.g., Proposition 5 of Foster and Krishnamurthy [2021]).

Since  $\hat{y}, y^* \in [0, 1]$ , their sum is at most 2 ( $\hat{y} + y^* \leq 2$ ). Rearranging the first inequality, we obtain

$$(\hat{y} - y^*)^2 \leq 2(\hat{y} + y^*)d_{\text{KL}}(\hat{y}, y^*) \leq 4d_{\text{KL}}(\hat{y}, y^*).$$

□

**Lemma 2.** Let  $C_1 = |\ell_{11} - \ell_{10}|$  be the constant for the Accept action ( $a = 1$ ). From the definition of  $\psi_1$ , it follows that:

$$(\psi_1(\hat{y}_t) - \psi_1(y_t^*))^2 = C_1^2(\hat{y}_t - y_t^*)^2. \quad (5)$$

Then, under the Assumption 2 and Freedman's inequality, with probability at least  $1 - \delta$ , the following bound holds:

$$\sum_{t=1}^T p_{t,1}(\psi_1(\hat{y}_t) - \psi_1(y_t^*))^2 \leq 8C_1^2 \text{Reg}_{\text{KL}}(T) + 4C_1^2 \log(2\delta^{-1}). \quad (6)$$

*Proof.* Let  $\mathcal{F}_{t-1} = \sigma((x_1, a_1, y_1), \dots, (x_{t-1}, a_{t-1}, y_{t-1}), x_t)$  be the filtration. Define

$$M_t := \mathbb{1}\{a_t = 1\}(\hat{y}_t - y_t^*)^2.$$

Set  $Z_t := M_t - \mathbb{E}[M_t \mid \mathcal{F}_{t-1}]$ . Since  $(\hat{y}_t - y_t^*)^2 \leq 1$  and  $0 \leq \mathbb{I}\{a_t = 1\} \leq 1$ , the range is  $0 \leq M_t \leq 1$ . Then,

$$\mathbb{E}[M_t \mid \mathcal{F}_{t-1}] = p_{t,1}(\hat{y}_t - y_t^*)^2, \quad \mathbb{E}[Z_t^2 \mid \mathcal{F}_{t-1}] \leq \mathbb{E}[M_t \mid \mathcal{F}_{t-1}].$$

Applying Freedman's inequality with range bound 1 and  $\eta = \frac{1}{2}$ , with probability at least  $1 - \delta$

$$\sum_{t=1}^T \mathbb{E}[M_t \mid \mathcal{F}_{t-1}] \leq 2 \sum_{t=1}^T M_t + 4 \log(2\delta^{-1}). \quad (7)$$

By Lemma 1, we know  $d_{\text{KL}}(\hat{y}_t, y_t^*) \geq \frac{1}{2} \frac{(\hat{y}_t - y_t^*)^2}{\hat{y}_t + y_t^*}$ . Since  $\hat{y}_t, y_t^* \in [0, 1]$ , we have  $\hat{y}_t + y_t^* \leq 2$ , which implies  $(\hat{y}_t - y_t^*)^2 \leq 4d_{\text{KL}}(\hat{y}_t, y_t^*)$ . Therefore, based on the Log-loss oracle assumption, we can bound the sum of  $M_t$ :

$$\sum_{t=1}^T M_t = \sum_{t=1}^T \mathbb{I}\{a_t = 1\}(\hat{y}_t - y_t^*)^2 \leq 4 \sum_{t=1}^T \mathbb{I}\{a_t = 1\}d_{\text{KL}}(\hat{y}_t, y_t^*) \leq 4\text{Reg}_{\text{KL}}(T).$$

Plugging this into (7), we get

$$\sum_{t=1}^T p_{t,1}(\hat{y}_t - y_t^*)^2 \leq 8\text{Reg}_{\text{KL}}(T) + 4 \log(2\delta^{-1}).$$

Finally, multiplying both sides by  $C_1^2$  and using  $(\psi_1(\hat{y}_t) - \psi_1(y_t^*))^2 = C_1^2(\hat{y}_t - y_t^*)^2$ , we obtain

$$\sum_{t=1}^T p_{t,1}(\psi_1(\hat{y}_t) - \psi_1(y_t^*))^2 \leq 8C_1^2\text{Reg}_{\text{KL}}(T) + 4C_1^2 \log(2\delta^{-1}).$$

□

**Lemma 3.** Suppose Assumptions 1 and 2 hold. Let  $\mathcal{A} = \{1, 0\}$  be the action set, where  $a = 1$  denotes the Accept action and  $a = 0$  denotes the Reject action. Then, with probability at least  $1 - \delta$ , the cumulative regret  $R(T)$  is bounded as

$$R(T) \leq \sum_{t=1}^T \sum_{a \in \mathcal{A}} p_{t,a} \left( \psi_a(y_t^*) - \psi_{a_t^*}(y_t^*) \right) + R' \sqrt{2T \log(2\delta^{-1})} \quad (8)$$

where  $R' = \max(\ell_{10}, \ell_{01} - \ell_{11})$ .

*Proof.* Let  $\mathcal{F}_{t-1} = \sigma((x_1, a_1, y_1), \dots, (x_{t-1}, a_{t-1}, y_{t-1}), x_t)$  be the filtration representing the history up to round  $t$ . The conditional expectation of the instantaneous regret  $r_t = \psi_{a_t}(y_t) - \psi_{a_t^*}(y_t)$  with respect to  $\mathcal{F}_{t-1}$  is given by

$$\begin{aligned} \mathbb{E}_{t-1}[r_t] &= \mathbb{E} \left[ \psi_{a_t}(y_t) - \psi_{a_t^*}(y_t) \mid \mathcal{F}_{t-1} \right] \\ &= \sum_{a \in \mathcal{A}} p_{t,a} \left( \psi_a(y_t^*) - \psi_{a_t^*}(y_t^*) \right). \end{aligned}$$

We define a martingale difference sequence  $Z_t$  as the deviation of the realized regret from its conditional expectation

$$Z_t = r_t - \mathbb{E}_{t-1}[r_t].$$

By construction,  $\mathbb{E}_{t-1}[Z_t] = 0$ . To apply Azuma-Hoeffding's inequality, we determine the range of  $r_t$  by considering the possible values of the true label  $y_t \in \{0, 1\}$ :

- **Case  $y_t = 0$ :** The losses are  $\psi_0(0) = 0$  (Reject) and  $\psi_1(0) = \ell_{10}$  (Accept). Since  $\ell_{10} \geq 0$  by Assumption 1, the optimal action is  $a_t^* = 0$ . Thus,  $r_t = \psi_{a_t}(y_t) - \psi_{a_t^*}(y_t) = \psi_{a_t}(0)$ . Since  $a_t \in \{0, 1\}$ , we have  $r_t \in \{0, \ell_{10}\}$ .

- **Case  $y_t = 1$ :** The losses are  $\psi_0(1) = \ell_{01}$  (Reject) and  $\psi_1(1) = \ell_{11}$  (Accept). Since  $\ell_{01} \geq \ell_{11}$  by Assumption 2, the optimal action is  $a_t^* = 1$ . Thus,  $r_t = \psi_{a_t}(1) - \psi_1(1)$ . If  $a_t = 1$ ,  $r_t = 0$ ; if  $a_t = 0$ ,  $r_t = \ell_{01} - \ell_{11}$ . This implies  $r_t \in \{0, \ell_{01} - \ell_{11}\}$ .

Combining both cases, the instantaneous regret is bounded within the interval  $r_t \in [0, \max(\ell_{10}, \ell_{01} - \ell_{11})]$ . Let  $R' = \max(\ell_{10}, \ell_{01} - \ell_{11})$  be this upper bound. Since  $0 \leq \mathbb{E}_{t-1}[r_t] \leq R'$ , the range of the martingale difference  $Z_t$  is also bounded by  $R'$ .

Applying Azuma-Hoeffding's inequality to the sum  $\sum_{t=1}^T Z_t$ , we have that with probability at least  $1 - \delta$

$$\sum_{t=1}^T (r_t - \mathbb{E}_{t-1}[r_t]) \leq R' \sqrt{2T \log(\delta^{-1})}.$$

we obtain

$$\begin{aligned} R(T) &= \sum_{t=1}^T r_t \leq \sum_{t=1}^T \mathbb{E}_{t-1}[r_t] + R' \sqrt{2T \log(2\delta^{-1})} \\ &= \sum_{t=1}^T \sum_{a \in \mathcal{A}} p_{t,a} \left( \psi_a(y_t^*) - \psi_{a_t^*}(y_t^*) \right) + R' \sqrt{2T \log(2\delta^{-1})}. \end{aligned}$$

This completes the proof.  $\square$

## A.2 Regret Decomposition

We partition the total rounds  $[T]$  into two disjoint sets based on the learner's greedy decision  $b_t = \arg \min_{a \in \{1,0\}} \psi_a(y_t)$ :

- **$B_1$  Rounds ( $\mathcal{I}_1$ ):**  $\{t \in [T] : b_t = 1\}$ .
- **$B_0$  Rounds ( $\mathcal{I}_0$ ):**  $\{t \in [T] : b_t = 0\}$ .

The regret in  $\mathcal{I}_0$  requires an additional refined decomposition to address the lack of direct feedback. Specifically, we further categorize the rounds in  $\mathcal{I}_0$  by comparing the greedy choice with the true optimal action  $a_t^*$ :

- $a_t^* = 1$  (Optimal is Accept): The optimal action is to Accept, but the greedy choice is to Reject. This represents a misclassification where the learner misses an opportunity to observe.
- $a_t^* = 0$  (Optimal is Reject): The optimal action is to Reject, and the greedy choice is correct. In this case, regret is only incurred due to the learner's stochastic exploration, specifically when the Accept action is sampled with probability  $p_{t,1}$ .

The resulting comprehensive decomposition of the cumulative regret  $R(T)$  is as follows

$$\begin{aligned} R(T) &= \sum_{t \in \mathcal{I}_1} r_t + \sum_{t \in \mathcal{I}_0} r_t \\ &= \sum_{t \in \mathcal{I}_1} r_t + \sum_{t \in \mathcal{I}_0, a_t^* = 1} r_t + \sum_{t \in \mathcal{I}_0, a_t^* = 0} r_t \\ &= R_1(T) + R_{0,1}(T) + R_{0,0}(T). \end{aligned} \tag{9}$$

By applying Lemma 3, the realized cumulative regret  $R(T)$  can be bounded by the sum of conditional expected regrets (denoted as  $\bar{R}$ ) and the martingale concentration term. Specifically, with probability at least  $1 - \delta$ :

$$R(T) \leq \bar{R}_1(T) + \bar{R}_{0,1}(T) + \bar{R}_{0,0}(T) + R' \sqrt{2T \log(2\delta^{-1})}, \tag{10}$$

where  $\bar{R}_1(T) = \sum_{t \in \mathcal{I}_1} \mathbb{E}_{t-1}[r_t]$ ,  $\bar{R}_{0,1}(T) = \sum_{t \in \mathcal{I}_0, a_t^* = 1} \mathbb{E}_{t-1}[r_t]$ , and  $\bar{R}_{0,0}(T) = \sum_{t \in \mathcal{I}_0, a_t^* = 0} \mathbb{E}_{t-1}[r_t]$ .

**Corollary 2.** Suppose Assumptions 1 and 2 hold. Following Lemma 3, with probability at least  $1 - \delta$ , the cumulative regret  $R(T)$  can be decomposed and bounded as

$$R(T) \leq \sum_{t \in \mathcal{I}_1} \sum_{a \in \mathcal{A}} p_{t,a} \left[ (\psi_a(y_t^*) - \psi_{a_t^*}(y_t^*)) - \frac{\gamma}{4} (\psi_a(\hat{y}_t) - \psi_a(y_t^*))^2 \right] \quad (11)$$

$$+ \frac{\gamma}{4} \sum_{t \in \mathcal{I}_1} \sum_{a \in \mathcal{A}} p_{t,a} (\psi_a(\hat{y}_t) - \psi_a(y_t^*))^2 \quad (12)$$

$$+ \bar{R}_0(T) + R' \sqrt{2T \log(2\delta^{-1})}. \quad (13)$$

*Proof.* From Lemma 3, with probability at least  $1 - \delta$ , the realized cumulative regret  $R(T)$  is bounded by the sum of conditional expected regrets and the martingale concentration term.

$$R(T) \leq \bar{R}_1(T) + \bar{R}_0(T) + R' \sqrt{2T \log(2\delta^{-1})},$$

where the conditional expected regrets are defined as  $\bar{R}_1(T) = \sum_{t \in \mathcal{I}_1} \sum_{a \in \mathcal{A}} p_{t,a} (\psi_a(y_t^*) - \psi_{a_t^*}(y_t^*))$  and  $\bar{R}_0(T) = \sum_{t \in \mathcal{I}_0} \sum_{a \in \mathcal{A}} p_{t,a} (\psi_a(y_t^*) - \psi_{a_t^*}(y_t^*))$ .

To connect the regret in  $B_1$  rounds ( $\bar{R}_1(T)$ ) to the regression oracle, we employ an additive-subtractive trick. We add and subtract the term  $\frac{\gamma}{4} (\psi_a(\hat{y}_t) - \psi_a(y_t^*))^2$  for each action in the  $\bar{R}_1(T)$  summation.

$$\begin{aligned} \bar{R}_1(T) &= \sum_{t \in \mathcal{I}_1} \sum_{a \in \mathcal{A}} p_{t,a} \left[ (\psi_a(y_t^*) - \psi_{a_t^*}(y_t^*)) - \frac{\gamma}{4} (\psi_a(\hat{y}_t) - \psi_a(y_t^*))^2 + \frac{\gamma}{4} (\psi_a(\hat{y}_t) - \psi_a(y_t^*))^2 \right] \\ &= \sum_{t \in \mathcal{I}_1} \sum_{a \in \mathcal{A}} p_{t,a} \left[ (\psi_a(y_t^*) - \psi_{a_t^*}(y_t^*)) - \frac{\gamma}{4} (\psi_a(\hat{y}_t) - \psi_a(y_t^*))^2 \right] \\ &\quad + \frac{\gamma}{4} \sum_{t \in \mathcal{I}_1} \sum_{a \in \mathcal{A}} p_{t,a} (\psi_a(\hat{y}_t) - \psi_a(y_t^*))^2. \end{aligned}$$

□

**Lemma 4.** Suppose Assumptions 1 and 2 hold. Let  $C_1 = |\ell_{11} - \ell_{10}|$ . By the definition of the loss function  $\psi$ , we have  $(\psi_0(\hat{y}_t) - \psi_0(y_t^*))^2 = \ell_{01}^2 (\hat{y}_t - y_t^*)^2$ . Then, with probability at least  $1 - \delta$ , the expected squared error of the loss estimates for rounds in  $\mathcal{I}_1$  is bounded as

$$\sum_{t \in \mathcal{I}_1} \sum_{a \in \{1,0\}} p_{t,a} (\psi_a(\hat{y}_t) - \psi_a(y_t^*))^2 \leq \Lambda_1 \left( 8\text{Reg}_{\text{KL}}(T) + 4 \log(2\delta^{-1}) \right), \quad (14)$$

where the constant  $\Lambda_1$  is defined as  $\Lambda_1 := C_1^2 + \frac{\ell_{01}^2}{\mu - 1}$ .

*Proof.* For rounds  $t \in \mathcal{I}_1$ , the greedy action is to Accept ( $b_t = 1$ ). We decompose the total squared error sum into components for each action  $a \in \{1, 0\}$

$$\sum_{t \in \mathcal{I}_1} \sum_{a \in \{1,0\}} p_{t,a} (\psi_a(\hat{y}_t) - \psi_a(y_t^*))^2 = \sum_{t \in \mathcal{I}_1} p_{t,1} (\psi_1(\hat{y}_t) - \psi_1(y_t^*))^2 + \sum_{t \in \mathcal{I}_1} p_{t,0} (\psi_0(\hat{y}_t) - \psi_0(y_t^*))^2. \quad (15)$$

For the first term (Accept action  $a = 1$ ), we can apply the bound derived from Freedman's inequality in Lemma 2, summing only over  $t \in \mathcal{I}_1 \subseteq [T]$

$$\sum_{t \in \mathcal{I}_1} p_{t,1} (\psi_1(\hat{y}_t) - \psi_1(y_t^*))^2 \leq \sum_{t=1}^T p_{t,1} (\psi_1(\hat{y}_t) - \psi_1(y_t^*))^2 \leq C_1^2 \left( 8\text{Reg}_{\text{KL}}(T) + 4 \log(2\delta^{-1}) \right). \quad (16)$$

For the second term (Reject action  $a = 0$ ), we relate its error to the Accept action by bounding the probability ratio. Since  $b_t = 1$  for  $t \in \mathcal{I}_1$ , we know that  $\psi_0(\hat{y}_t) > \psi_1(\hat{y}_t)$ . Based on the SquareCB



sampling rule, the probabilities are defined as

$$p_{t,0} = \frac{1}{\mu + \gamma(\psi_0(\hat{y}_t) - \psi_1(\hat{y}_t))}, \quad p_{t,1} = 1 - p_{t,0}.$$

The probability ratio is then:

$$\frac{p_{t,0}}{p_{t,1}} = \frac{p_{t,0}}{1 - p_{t,0}} = \frac{1}{\mu + \gamma(\psi_0(\hat{y}_t) - \psi_1(\hat{y}_t)) - 1}.$$

Since this ratio is decreasing with respect to the difference  $\psi_0(\hat{y}_t) - \psi_1(\hat{y}_t)$  (which is strictly positive), the maximum occurs as the difference approaches 0. Thus, we have the upper bound

$$\frac{p_{t,0}}{p_{t,1}} < \frac{1}{\mu - 1}.$$

Using the identity  $(\psi_0(\hat{y}_t) - \psi_0(y_t^*))^2 = \frac{\ell_{01}^2}{C_1^2} (\psi_1(\hat{y}_t) - \psi_1(y_t^*))^2$ , we relate the squared error terms

$$\begin{aligned} p_{t,0}(\psi_0(\hat{y}_t) - \psi_0(y_t^*))^2 &= \frac{p_{t,0}}{p_{t,1}} \cdot \frac{\ell_{01}^2}{C_1^2} \cdot p_{t,1}(\psi_1(\hat{y}_t) - \psi_1(y_t^*))^2 \\ &< \frac{1}{\mu - 1} \cdot \frac{\ell_{01}^2}{C_1^2} \cdot p_{t,1}(\psi_1(\hat{y}_t) - \psi_1(y_t^*))^2. \end{aligned}$$

Summing this over all  $t \in \mathcal{I}_1$  and applying Lemma 2 yields

$$\begin{aligned} \sum_{t \in \mathcal{I}_1} p_{t,0}(\psi_0(\hat{y}_t) - \psi_0(y_t^*))^2 &\leq \frac{1}{\mu - 1} \frac{\ell_{01}^2}{C_1^2} \sum_{t=1}^T p_{t,1}(\psi_1(\hat{y}_t) - \psi_1(y_t^*))^2 \\ &\leq \frac{\ell_{01}^2}{\mu - 1} \left( 8\text{Reg}_{\text{KL}}(T) + 4\log(2\delta^{-1}) \right). \end{aligned} \quad (17)$$

Substituting (16) and (17) back into (15), we arrive at the final bound

$$\begin{aligned} \sum_{t \in \mathcal{I}_1} \sum_{a \in \{1,0\}} p_{t,a}(\psi_a(\hat{y}_t) - \psi_a(y_t^*))^2 &\leq \left( C_1^2 + \frac{\ell_{01}^2}{\mu - 1} \right) \left( 8\text{Reg}_{\text{KL}}(T) + 4\log(2\delta^{-1}) \right) \\ &= \Lambda_1 \left( 8\text{Reg}_{\text{KL}}(T) + 4\log(2\delta^{-1}) \right). \end{aligned}$$

This completes the proof.  $\square$

**Lemma 5.** Suppose Assumptions 1 and 2 hold. Let  $\mathcal{I}_0 = \{t \in [T] : b_t = 0\}$  be the set of rounds where the greedy choice is the Reject action. Define the loss gap between the Reject and Accept actions as  $\Delta(y) := \psi_0(y) - \psi_1(y)$ , and let  $C_\Delta := |\ell_{01} - \ell_{11} + \ell_{10}|$ . By the definition of the linear loss functions, we have  $|\Delta(y) - \Delta(y')| = C_\Delta |y - y'|$  for all  $y, y' \in [0, 1]$ . For the SquareCB exploration parameter  $\mu = 2$ , with probability at least  $1 - \delta$ , the sum of conditional expected regrets in  $B_0$  rounds,  $\bar{R}_0(T) = \bar{R}_{0,1}(T) + \bar{R}_{0,0}(T)$ , is bounded by

$$\begin{aligned} \bar{R}_0(T) &\leq \frac{T}{\gamma} + 2C_\Delta \sqrt{T(8\text{Reg}_{\text{KL}}(T) + 4\log(2\delta^{-1}))} \\ &\quad + \gamma C_\Delta^2 \left( 8\text{Reg}_{\text{KL}}(T) + 4\log(2\delta^{-1}) \right). \end{aligned} \quad (18)$$

*Proof.* For any round  $t \in \mathcal{I}_0$ , the greedy choice is  $b_t = 0$ , which implies that the estimated gap is non-positive  $\hat{\Delta}_t := \Delta(\hat{y}_t) \leq 0$ . The true gap is denoted as  $\Delta_t^* := \Delta(y_t^*)$ . According to the LogCB sampling rule, the exploration probability for the Accept action ( $a = 1$ ) is given by  $p_{t,1} = \frac{1}{2 + \gamma(\psi_1(\hat{y}_t) - \psi_0(\hat{y}_t))} = \frac{1}{2 + \gamma(-\hat{\Delta}_t)}$ .

We analyze the conditional expected regret  $\mathbb{E}_{t-1}[r_t]$  for  $t \in \mathcal{I}_0$ . By definition, the conditional expected regret is the sum of instantaneous regrets over all possible sampled actions  $a \in \{1, 0\}$ , weighted by their exploration probabilities  $p_{t,a}$ :

$$\mathbb{E}_{t-1}[r_t] = \sum_{a \in \{1, 0\}} p_{t,a} (\psi_a(y_t^*) - \psi_{a_t^*}(y_t^*)). \quad (19)$$

Notice that when the sampled action matches the optimal action ( $a = a_t^*$ ), the term becomes zero (e.g.,  $\psi_1(y_t^*) - \psi_1(y_t^*) = 0$ ). Thus, regret is strictly incurred only when the algorithm explores a suboptimal action ( $a \neq a_t^*$ ). We consider two exhaustive cases based on the true optimal action  $a_t^* \in \{1, 0\}$ .

**Case 1:**  $a_t^* = 1$  ( $\bar{R}_{0,1}$ ) The optimal action is Accept, meaning the true gap is positive ( $\Delta_t^* > 0$ ). Expanding the expected regret definition from (19), the regret for sampling the correct action ( $a = 1$ ) vanishes:

$$\begin{aligned} \mathbb{E}_{t-1}[r_t] &= p_{t,1}(\psi_1(y_t^*) - \psi_1(y_t^*)) + p_{t,0}(\psi_0(y_t^*) - \psi_1(y_t^*)) \\ &= 0 + p_{t,0}\Delta_t^* = p_{t,0}\Delta_t^*. \end{aligned}$$

Using the fact that  $p_{t,0} = 1 - p_{t,1}$ , the probability ratio is

$$\frac{p_{t,0}}{p_{t,1}} = \frac{1 - p_{t,1}}{p_{t,1}} = \left(2 + \gamma(-\hat{\Delta}_t)\right) - 1 = 1 + \gamma(-\hat{\Delta}_t).$$

We can rewrite the expected regret as

$$\mathbb{E}_{t-1}[r_t] = p_{t,1} \frac{p_{t,0}}{p_{t,1}} \Delta_t^* = p_{t,1}(1 + \gamma(-\hat{\Delta}_t))\Delta_t^*.$$

Since  $\Delta_t^* > 0$  and  $\hat{\Delta}_t \leq 0$ , we bound the terms using the definition of  $\Delta$ :

- $-\hat{\Delta}_t \leq \Delta_t^* - \hat{\Delta}_t \leq |\Delta_t^* - \hat{\Delta}_t| = C_\Delta |y_t^* - \hat{y}_t|$ .
- $\Delta_t^* = \hat{\Delta}_t + (\Delta_t^* - \hat{\Delta}_t) \leq \Delta_t^* - \hat{\Delta}_t \leq |\Delta_t^* - \hat{\Delta}_t| = C_\Delta |y_t^* - \hat{y}_t|$  (because  $\hat{\Delta}_t \leq 0$ ).

Substituting these upper bounds yields:

$$\mathbb{E}_{t-1}[r_t] \leq p_{t,1}(1 + \gamma C_\Delta |y_t^* - \hat{y}_t|) C_\Delta |y_t^* - \hat{y}_t| = C_\Delta p_{t,1} |y_t^* - \hat{y}_t| + \gamma C_\Delta^2 p_{t,1} (y_t^* - \hat{y}_t)^2.$$

**Case 2:**  $a_t^* = 0$  ( $\bar{R}_{0,0}$ ) The optimal action is Reject, meaning  $\Delta_t^* \leq 0$ . Even though the greedy action is correct ( $b_t = a_t^* = 0$ ), regret is incurred due to stochastic exploration when  $a = 1$  is sampled. Expanding the expected regret, the term for sampling the correct action ( $a = 0$ ) vanishes:

$$\begin{aligned} \mathbb{E}_{t-1}[r_t] &= p_{t,1}(\psi_1(y_t^*) - \psi_0(y_t^*)) + p_{t,0}(\psi_0(y_t^*) - \psi_0(y_t^*)) \\ &= p_{t,1}(-\Delta_t^*) + 0 = p_{t,1}(-\Delta_t^*). \end{aligned}$$

We decompose  $-\Delta_t^*$  by adding and subtracting  $\hat{\Delta}_t$ :

$$-\Delta_t^* = -\hat{\Delta}_t + (\hat{\Delta}_t - \Delta_t^*) \leq -\hat{\Delta}_t + |\hat{\Delta}_t - \Delta_t^*| = -\hat{\Delta}_t + C_\Delta |\hat{y}_t - y_t^*|.$$

Multiplying by  $p_{t,1}$ , we obtain:

$$\mathbb{E}_{t-1}[r_t] \leq p_{t,1}(-\hat{\Delta}_t) + C_\Delta p_{t,1} |\hat{y}_t - y_t^*|.$$

Notice that  $p_{t,1}(-\hat{\Delta}_t) = \frac{-\hat{\Delta}_t}{2 + \gamma(-\hat{\Delta}_t)}$ . Since the function  $f(x) = \frac{x}{2 + \gamma x}$  is monotonically increasing for  $x \geq 0$  and bounded by  $\frac{1}{\gamma}$ , and given  $-\hat{\Delta}_t \geq 0$ , we have  $p_{t,1}(-\hat{\Delta}_t) \leq \frac{1}{\gamma}$ . Therefore,

$$\mathbb{E}_{t-1}[r_t] \leq \frac{1}{\gamma} + C_\Delta p_{t,1} |\hat{y}_t - y_t^*|.$$

**Combining the Bounds** Summing the expected regrets from both cases over all rounds  $t \in \mathcal{I}_0$ , and relaxing the summation to all  $T$  rounds since  $\mathcal{I}_0 \subseteq [T]$  and all terms are non-negative, we get:

$$\begin{aligned}\bar{R}_0(T) &= \sum_{t \in \mathcal{I}_0} \mathbb{E}_{t-1}[r_t] \leq \sum_{t \in \mathcal{I}_0} \left[ \frac{1}{\gamma} + 2C_{\Delta} p_{t,1} |y_t^* - \hat{y}_t| + \gamma C_{\Delta}^2 p_{t,1} (y_t^* - \hat{y}_t)^2 \right] \\ &\leq \frac{T}{\gamma} + 2C_{\Delta} \sum_{t=1}^T p_{t,1} |y_t^* - \hat{y}_t| + \gamma C_{\Delta}^2 \sum_{t=1}^T p_{t,1} (y_t^* - \hat{y}_t)^2.\end{aligned}$$

Applying the Cauchy-Schwarz inequality to the middle term:

$$\sum_{t=1}^T p_{t,1} |y_t^* - \hat{y}_t| = \sum_{t=1}^T \sqrt{p_{t,1}} \cdot \sqrt{p_{t,1}} |y_t^* - \hat{y}_t| \leq \sqrt{\sum_{t=1}^T p_{t,1}} \sqrt{\sum_{t=1}^T p_{t,1} (y_t^* - \hat{y}_t)^2} \leq \sqrt{T \sum_{t=1}^T p_{t,1} (y_t^* - \hat{y}_t)^2}.$$

Substituting this back into the expected regret bound yields:

$$\bar{R}_0(T) \leq \frac{T}{\gamma} + 2C_{\Delta} \sqrt{T \sum_{t=1}^T p_{t,1} (y_t^* - \hat{y}_t)^2} + \gamma C_{\Delta}^2 \sum_{t=1}^T p_{t,1} (y_t^* - \hat{y}_t)^2.$$

From Lemma 2, we know that with probability at least  $1 - \delta$ , the sum of squared errors is bounded by  $\sum_{t=1}^T p_{t,1} (\hat{y}_t - y_t^*)^2 \leq 8\text{Reg}_{\text{KL}}(T) + 4\log(2\delta^{-1})$ . Applying this upper bound yields the final result.  $\square$

**Lemma 6** (Lemma 3 of SquareCB[Foster and Rakhlin, 2020]). *For any round  $t$ , the probability distribution  $p_t$  over the action set  $\mathcal{A} = \{1, 0\}$  (where  $K = 2$ ), chosen by the SquareCB algorithm, ensures that for any true loss vector defined by  $y_t^*$ :*

$$\sum_{a \in \mathcal{A}} p_{t,a} \left[ (\psi_a(y_t^*) - \psi_{a_t^*}(y_t^*)) - \frac{\gamma}{4} (\psi_a(\hat{y}_t) - \psi_a(y_t^*))^2 \right] \leq \frac{4}{\gamma}. \quad (20)$$

Summing this inequality over all rounds  $t \in \mathcal{I}_1$ , we obtain:

$$\sum_{t \in \mathcal{I}_1} \sum_{a \in \mathcal{A}} p_{t,a} \left[ (\psi_a(y_t^*) - \psi_{a_t^*}(y_t^*)) - \frac{\gamma}{4} (\psi_a(\hat{y}_t) - \psi_a(y_t^*))^2 \right] \leq \frac{4|\mathcal{I}_1|}{\gamma} \leq \frac{4T}{\gamma}. \quad (21)$$

*Proof.* The first inequality is a direct application of Lemma 3 in SquareCB[Foster and Rakhlin, 2020], adapted to our binary action setting with  $K = 2$ . By substituting the arbitrary loss  $f_a^*$  with our specific loss function  $\psi_a(y_t^*)$  and the estimator  $\hat{y}_a$  with  $\psi_a(\hat{y}_t)$ , the bound holds for each individual round  $t$ . Summing this bound over all rounds  $t \in \mathcal{I}_1$  and using the trivial upper bound  $|\mathcal{I}_1| \leq T$  yields the final cumulative result.  $\square$

**Theorem 3** (LogCB-AT Regret Bound). *Suppose Assumptions 1 and 2 hold. For the choice of  $\gamma = O(\sqrt{T/\text{Reg}_{\text{KL}}(T)})$ , the cumulative regret of the LogCB-AT algorithm is bounded by:*

$$R(T) = O\left(\sqrt{T \cdot \text{Reg}_{\text{KL}}(T)}\right)$$

*with probability at least  $1 - \delta$ .*

*Proof.* From Corollary 2,

$$\begin{aligned}R(T) &\leq \sum_{t \in \mathcal{I}_1} \sum_{a \in \{1,0\}} p_{t,a} \left[ (\psi_a(y_t^*) - \psi_{a_t^*}(y_t^*)) - \frac{\gamma}{4} (\psi_a(\hat{y}_t) - \psi_a(y_t^*))^2 \right] \\ &\quad + \frac{\gamma}{4} \sum_{t \in \mathcal{I}_1} \sum_{a \in \{1,0\}} p_{t,a} (\psi_a(\hat{y}_t) - \psi_a(y_t^*))^2 \\ &\quad + \bar{R}_0(T) + R' \sqrt{2T \log(2\delta^{-1})}.\end{aligned}$$

By substituting the upper bounds derived in Lemma 4, Lemma 5, and Lemma 6 into the inequality, we obtain

$$\begin{aligned} R(T) &\leq \frac{4T}{\gamma} + \frac{\gamma}{4}\Lambda_1 \left(8\text{Reg}_{\text{KL}}(T) + 4\log(2\delta^{-1})\right) \\ &\quad + \frac{T}{\gamma} + 2C_\Delta \sqrt{T \cdot (8\text{Reg}_{\text{KL}}(T) + 4\log(2\delta^{-1}))} + \gamma C_\Delta^2 \left(8\text{Reg}_{\text{KL}}(T) + 4\log(2\delta^{-1})\right) \\ &\quad + R' \sqrt{2T \log(2\delta^{-1})}. \end{aligned}$$

Grouping the terms by  $\frac{1}{\gamma}$  and  $\gamma$ , we have:

$$\begin{aligned} R(T) &\leq \frac{5T}{\gamma} + \gamma \left[ (2\Lambda_1 + 8C_\Delta^2) \text{Reg}_{\text{KL}}(T) + (\Lambda_1 + 4C_\Delta^2) \log(2\delta^{-1}) \right] \\ &\quad + 2C_\Delta \sqrt{T \cdot (8\text{Reg}_{\text{KL}}(T) + 4\log(2\delta^{-1}))} + R' \sqrt{2T \log(2\delta^{-1})}. \end{aligned}$$

By setting the learning rate  $\gamma$  to balance the first two terms:

$$\gamma = \sqrt{\frac{5T}{(2\Lambda_1 + 8C_\Delta^2) \text{Reg}_{\text{KL}}(T) + (\Lambda_1 + 4C_\Delta^2) \log(2\delta^{-1})}}, \quad (22)$$

the sum of the  $\frac{1}{\gamma}$  and  $\gamma$  terms simplifies to  $2\sqrt{5T \left[ (2\Lambda_1 + 8C_\Delta^2) \text{Reg}_{\text{KL}}(T) + (\Lambda_1 + 4C_\Delta^2) \log(2\delta^{-1}) \right]}$ .

This yields the final bounded expression:

$$\begin{aligned} R(T) &\leq 2\sqrt{5T \left[ (2\Lambda_1 + 8C_\Delta^2) \text{Reg}_{\text{KL}}(T) + (\Lambda_1 + 4C_\Delta^2) \log(2\delta^{-1}) \right]} \\ &\quad + 2C_\Delta \sqrt{T \cdot (8\text{Reg}_{\text{KL}}(T) + 4\log(2\delta^{-1}))} + R' \sqrt{2T \log(2\delta^{-1})}. \end{aligned}$$

Extracting the dominant terms with respect to  $T$ , we conclude

$$R(T) = O\left(\sqrt{T \cdot \text{Reg}_{\text{KL}}(T)}\right). \quad (23)$$

□

## B SquareCB-AT Regret Analysis

In this section, we present the regret analysis for the Contextual Apple Tasting problem under the square loss setting, serving as an alternative to the log-loss framework used in LogCB-AT. To ensure compatibility with the standard SquareCB architecture, which inherently relies on square loss regression oracles, we adopt a **Linear Probability Model (LPM)** as our generative foundation. Under this setting, the conditional expectation is modeled linearly with a fixed offset:

$$y_t \mid x_t \sim \text{Bernoulli}\left(x_t^\top \theta^* + \frac{1}{2}\right). \quad (24)$$

Accordingly, we introduce our new algorithm (Algorithm 3) and assumption (Assumption 3).

**Assumption 3** (Squared-loss oracle guarantee). *For any (possibly adaptive) sequence of contexts, actions, and labels, the regression oracle satisfies*

$$\sum_{t=1}^T \mathbf{1}\{a_t = 1\} (\hat{y}_t - f^*(x_t))^2 \leq \text{Reg}_{\text{Sq}}(T_1) \leq \text{Reg}_{\text{Sq}}(T),$$

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**Algorithm 3** SquareCB-AT

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**Require:** Learning rate  $\gamma > 0$ , exploration parameter  $\mu > 1$ , online regression oracle  $\text{Alg}_{\text{Sq}}$

```
1: for  $t = 1, \dots, T$  do
2:   Observe context  $x_t \in \mathcal{X}$ 
3:   Obtain prediction  $\hat{y}_t = \hat{f}_{t-1}(x_t)$  from  $\text{SqAlg}$ 
4:   Compute  $\psi_0(\hat{y}_t)$  and  $\psi_1(\hat{y}_t)$ 
5:   Set
      
$$b_t \in \arg \min_{a \in \{0,1\}} \psi_a(\hat{y}_t)$$

6:   for each  $a \neq b_t$  do
7:     Set
      
$$p_{t,a} = \frac{1}{\mu + \gamma(\psi_a(\hat{y}_t) - \psi_{b_t}(\hat{y}_t))}$$

8:   end for
9:   Set
      
$$p_{t,b_t} = 1 - \sum_{a \neq b_t} p_{t,a}$$

10:  Sample  $a_t \sim p_t$ 
11:  if  $a_t = 1$  then
12:    Observe  $y_t$  and update  $\text{Alg}_{\text{Sq}}$  with  $(x_t, y_t)$ 
13:  end if
14: end for
```

---

where

$$T_1 := \sum_{t=1}^T \mathbf{1}\{a_t = 1\}$$

denotes the number of rounds on which the action  $A$  is taken.

Next, we introduce Lemma 7. This lemma serves a similar role to Lemma 2 in LogCB-AT.

**Lemma 7.** Let  $C_1 = |\ell_{11} - \ell_{10}|$  be the constant for the Accept action ( $a = 1$ ). From the definition of  $\psi_1$ , it follows that:

$$(\psi_1(\hat{y}_t) - \psi_1(y_t^*))^2 = C_1^2(\hat{y}_t - y_t^*)^2. \quad (25)$$

Then, under Assumption 2 and Freedman's inequality, with probability at least  $1 - \delta$ , the following bound holds:

$$\sum_{t=1}^T p_{t,1}(\psi_1(\hat{y}_t) - \psi_1(y_t^*))^2 \leq 2C_1^2 \text{Reg}_{\text{Sq}}(T) + 4C_1^2 \log(2\delta^{-1}). \quad (26)$$

*Proof.* Let  $\mathcal{F}_{t-1} = \sigma((x_1, a_1, y_1), \dots, (x_{t-1}, a_{t-1}, y_{t-1}), x_t)$  be the filtration. Define

$$M_t := \mathbf{1}\{a_t = 1\}(\hat{y}_t - y_t^*)^2.$$

Set  $Z_t := M_t - \mathbb{E}[M_t \mid \mathcal{F}_{t-1}]$ . Since  $(\hat{y}_t - y_t^*)^2 \leq 1$  and  $0 \leq \mathbf{1}\{a_t = 1\} \leq 1$ , the range is  $0 \leq M_t \leq 1$ . Then,

$$\mathbb{E}[M_t \mid \mathcal{F}_{t-1}] = p_{t,1}(\hat{y}_t - y_t^*)^2, \quad \mathbb{E}[Z_t^2 \mid \mathcal{F}_{t-1}] \leq \mathbb{E}[M_t \mid \mathcal{F}_{t-1}].$$

Applying Freedman's inequality with range bound 1 and  $\eta = \frac{1}{2}$ , with probability at least  $1 - \delta$ :

$$\sum_{t=1}^T \mathbb{E}[M_t \mid \mathcal{F}_{t-1}] \leq 2 \sum_{t=1}^T M_t + 4 \log(2\delta^{-1}). \quad (27)$$

By Assumption 2,  $\sum_{t=1}^T M_t = \sum_{t=1}^T \mathbb{1}\{a_t = 1\}(\hat{y}_t - y_t^*)^2 \leq \text{Reg}_{\text{Sq}}(T)$ . Plugging this into (27)

$$\sum_{t=1}^T p_{t,1}(\hat{y}_t - y_t^*)^2 \leq 2\text{Reg}_{\text{Sq}}(T) + 4\log(2\delta^{-1}).$$

Finally, multiplying both sides by  $C_1^2$  and using  $(\psi_1(\hat{y}_t) - \psi_1(y_t^*))^2 = C_1^2(\hat{y}_t - y_t^*)^2$ , we obtain

$$\sum_{t=1}^T p_{t,1}(\psi_1(\hat{y}_t) - \psi_1(y_t^*))^2 \leq 2C_1^2\text{Reg}_{\text{Sq}}(T) + 4C_1^2\log(2\delta^{-1}).$$

□

**Theorem 4** (SquareCB-AT Regret Bound). *Suppose Assumptions 1 and 3 hold. Then, with probability at least  $1 - \delta$ , the cumulative regret of the SquareCB-AT algorithm is bounded by:*

$$R(T) = O\left(\sqrt{T \cdot \text{Reg}_{\text{Sq}}(T)}\right). \quad (28)$$

*Proof.* The proof structure is essentially identical to the regret analysis of LogCB-AT presented in Theorem 3. The core steps and the regret decomposition remain exactly the same, with the only modification being the substitution of Lemma 2 (used for the log-loss setting) with Lemma 7 (derived for the square-loss setting).

we configure the optimal learning rate  $\gamma$  as

$$\gamma = \sqrt{\frac{5T}{\left(\frac{\Lambda_1}{2} + 2C_\Delta^2\right)\text{Reg}_{\text{Sq}}(T) + (\Lambda_1 + 4C_\Delta^2)\log(2\delta^{-1})}}. \quad (29)$$

Substituting this  $\gamma$  simplifies the sum of the  $\frac{1}{\gamma}$  and  $\gamma$  terms, yielding the explicit bounded expression:

$$\begin{aligned} R(T) &\leq 2\sqrt{5T \left[ \left(\frac{\Lambda_1}{2} + 2C_\Delta^2\right)\text{Reg}_{\text{Sq}}(T) + (\Lambda_1 + 4C_\Delta^2)\log(2\delta^{-1}) \right]} \\ &\quad + 2C_\Delta\sqrt{T \cdot \left(2\text{Reg}_{\text{Sq}}(T) + 4\log(2\delta^{-1})\right)} + R'\sqrt{2T\log(2\delta^{-1})}. \end{aligned} \quad (30)$$

Finally, we conclude

$$R(T) = O\left(\sqrt{T \cdot \text{Reg}_{\text{Sq}}(T)}\right). \quad (31)$$

□

## C LogCBPSide-AT Regret Analysis

In this section we provide the detailed proof for Theorem 1.

Our reward model is,

$$\mathbb{E}[y_t | x_t] = \sigma(x_t^T \cdot \theta^*)$$

Following the work of Filippi et al. [2010], we use *maximum quasi-likelihood estimator* for  $\hat{\theta}_t$ . That is,

$$\sum_{s=1}^t \mathbb{1}\{a_s = 1\} (y_s - \sigma(x_s^T \hat{\theta}_t)) x_s = 0.$$

Since, obtained rewards  $y_s$  are conditionally independent of the context ( $x_s$ ), we can modify the Proposition 1 of Filippi et al. [2010] from Appendix A.2 by replacing  $t$  with  $N_{t-1}$  (number of times action Reject was chosen).

**Proposition 1.** Take any  $\delta_t, N_{t-1}$  such that  $0 < \delta_t < \min\{1, \frac{d}{e}\}, 1 \leq N_{t-1} \leq T$ . Let  $x_t$  be any random variable. Let

$$\beta_{t-1}^{x_t}(\delta_t) = \frac{2k_\sigma R_{\max}}{c_\sigma} \|x_t\|_{V_{t-1}^{-1}} \sqrt{\left(3 + 2 \log\left(1 + \frac{2}{\lambda}\right)\right) \cdot 2d \log(N_{t-1}) \cdot \log\left(\frac{d}{\delta_t}\right)}.$$

Then, with probability at least  $1 - \delta_t$ , it holds that

$$|\sigma(x_t^T \theta^*) - \sigma(x_t^T \hat{\theta}_{t-1})| \leq \beta_{t-1}^{x_t}(\delta_t).$$

where,

- $R_{\max} := \max_t y_t = 1$ .
- $k_\sigma := \text{Lipschitz constant of } \sigma$ .

$$\begin{aligned} \sigma'(z) &= \sigma(z)(1 - \sigma(z)) \\ &\leq \frac{1}{4} \\ k_\sigma &= \frac{1}{4}. \end{aligned}$$

- $C_{\max} = \sup_{x_t \in \mathcal{X}, \theta^* \in \Theta} x_t^T \cdot \theta^*$
- $c_\sigma := \inf \sigma'(z)$ .

$$\begin{aligned} c_\sigma &= \inf \sigma(z)(1 - \sigma(z)) \\ &= \sup(|\sigma(z)|)(1 - \sup(|\sigma(z)|)) \\ &= \frac{e^{C_{\max}}}{(1 + e^{C_{\max}})^2}. \end{aligned}$$

Furthermore, we introduce 2 more Lemmas necessary for the regret analysis.

The Elliptical Potential Lemma (EPL),

**Lemma 8** (Elliptical potential lemma adapted from Proposition 2 of Abeille and Lazaric [2017]). Let  $x_1, x_2, \dots, x_t \in \mathbb{R}^d$  be a sequence of vectors with  $\|x_s\|_2 \leq 1, \forall s \in [t]$ . Let

$V_{t-1} = \lambda I + \sum_{s=1}^{t-1} x_s x_s^T$  for some  $\lambda > 0$ . Then,

$$\sum_{s=1}^t \|x_s\|_{V_{s-1}^{-1}}^2 \leq 2d \log(1 + \frac{t}{d\lambda}).$$

Adapting this to our case:

Let  $x_1 \cdot \mathbb{1}\{a_1 = 1\}, x_2 \cdot \mathbb{1}\{a_2 = 1\}, \dots, x_t \cdot \mathbb{1}\{a_t = 1\} \in \mathbb{R}^d$  for some  $\{a_s\}_{s=1}^t \in \{0, 1\}$ . Let  $V_{t-1} = \lambda I + \sum_{s=1}^{t-1} x_s x_s^T \mathbb{1}\{a_s = 1\}$  for some  $\lambda > 0$ . Then,

$$\sum_{s=1}^t \|x_s\|_{V_{s-1}^{-1}}^2 \mathbb{1}\{a_s = 1\} \leq 2d \log(1 + \frac{t}{d\lambda}).$$

and Elliptical Potential Count Lemma (EPC)

**Lemma 9** (Elliptical potential count lemma adapted from Lemma C.2 of Jun and Kim [2024]). Let  $x_1, x_2, \dots, x_t \in \mathbb{R}^d$  be a sequence of vectors with  $\|x_s\|_2 \leq 1, \forall s \in [t]$ . Let  $V_{t-1} = \lambda I + \sum_{s=1}^{t-1} x_s x_s^T$  for some  $\lambda > 0$ . Let  $J = \{s \in [t] : \|x_s\|_{V_{s-1}^{-1}}^2 \geq L^2\}$  for some  $L^2 \leq 1$ . Then,

$$|J| \leq 3 \frac{d}{L^2} \ln \left( 1 + \frac{2}{L^2 \lambda} \right).$$

Adapting this to our case:

Let  $x_1 \cdot \mathbb{1}\{a_1 = 1\}, x_2 \cdot \mathbb{1}\{a_2 = 1\}, \dots, x_t \cdot \mathbb{1}\{a_t = 1\} \in \mathbb{R}^d$  for some  $\{a_s\}_{s=1}^t \in \{0, 1\}$ . Let  $V_{t-1} = \lambda I + \sum_{s=1}^{t-1} x_s x_s^T \mathbb{1}\{a_s = 1\}$  for some  $\lambda > 0$ . Let  $J = \{s \in [t] : \|x_s \mathbb{1}\{a_s = 1\}\|_{V_{s-1}^{-1}}^2 \geq L^2\}$  for some  $L^2 \leq 1$ . Then,

$$|J| \leq 3 \frac{d}{L^2} \ln \left( 1 + \frac{2}{L^2 \lambda} \right).$$

### C.1 Minimax regret analysis

**Theorem 1** (LogCBPSide-AT minimax regret bound). For  $\delta_t = \frac{1}{t}$ , with  $\forall t \quad \|x_t\|_2 \leq 1$  and  $\sup_{x_t \in \mathcal{X}} x_t^T \theta^* \leq C_{\max}$ , the expected cumulative regret of the LogCBPSide-AT algorithm is bounded as:

$$\mathbb{E} [\text{Reg}_T] \leq O \left( \frac{(1 + e^{C_{\max}})^2}{e^{C_{\max}}} d \sqrt{T} \log^{\frac{3}{2}}(T) \right).$$

(Ignoring logarithmic terms in  $d$ )

Throughout the analysis, we use  $\hat{y}_t = \sigma(x_t^\top \hat{\theta}_{t-1})$  and  $y_t^* = \sigma(x_t^\top \theta^*)$  interchangeably to denote the estimated and true latent labels, respectively.

Let us define the good event  $\mathcal{G}_t$  as,

$$\mathcal{G}_t = \left\{ |\sigma(x_t^T \theta^*) - \sigma(x_t^T \hat{\theta}_{t-1})| \leq \beta_{t-1}^{x_t}(\delta_t) \right\}$$

Instantaneous regret,

$$r_t = \psi_{a_t}(y_t) - \psi_{a_t^*}(y_t)$$



Let  $\mathcal{F}_{t-1} = \sigma((x_1, a_1, y_1), \dots, (x_{t-1}, a_{t-1}, y_{t-1}), x_t)$  be the filtration representing the history up to round  $t$ . Furthermore, use the shorthand  $\mathbb{E}_{t-1}[\cdot]$  to denote  $\mathbb{E}[\cdot \mid \mathcal{F}_{t-1}]$ .

$$\mathbb{E}_{t-1}[r_t] = \psi_{a_t}(y_t^*) - \psi_{a_t^*}(y_t^*)$$

Cumulative regret,

$$\begin{aligned} \text{Reg}_T &= \sum_{t=1}^T r_t \\ \mathbb{E}[\text{Reg}_T] &= \mathbb{E}\left[\sum_{t=1}^T r_t\right] \\ \mathbb{E}[\text{Reg}_T] &= \underbrace{\mathbb{E}\left[\sum_{t=1}^T r_t \mathbb{1}\{\mathcal{G}_t\}\right]}_{F_1} + \underbrace{\mathbb{E}\left[\sum_{t=1}^T r_t \mathbb{1}\{\bar{\mathcal{G}}_t\}\right]}_{F_2} \end{aligned}$$

First, let us bound  $F_2$

$$\begin{aligned} F_2 &= \mathbb{E}\left[\sum_{t=1}^T r_t \mathbb{1}\{\bar{\mathcal{G}}_t\}\right] \\ &= \max(\ell_{10}, \ell_{01} - \ell_{11}) \sum_{t=1}^T \mathbb{P}(\bar{\mathcal{G}}_t) \\ &= \max(\ell_{10}, \ell_{01} - \ell_{11}) \sum_{t=1}^T \delta_t \\ &= \max(\ell_{10}, \ell_{01} - \ell_{11}) \sum_{t=1}^T \frac{1}{t} \quad (\text{Proposition 1, set } \delta_t = \frac{1}{t}) \\ &\lesssim \max(\ell_{10}, \ell_{01} - \ell_{11}) \log T \\ &= O(\log T) \end{aligned}$$

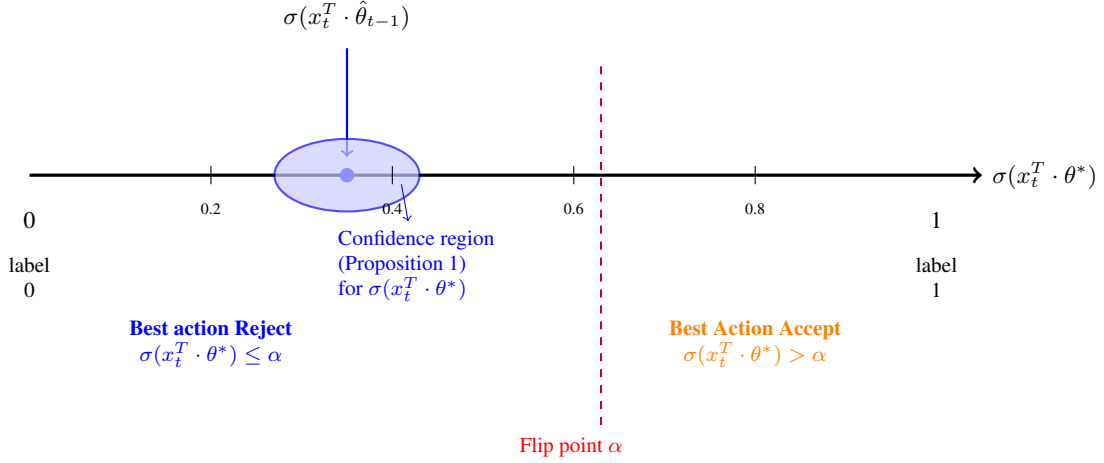
Now let us bound  $F_1$ ,

$$\begin{aligned} F_1 &= \sum_{t=1}^T \mathbb{E}[r_t \mathbb{1}\{\mathcal{G}_t\}] \\ &= \underbrace{\sum_{t=1}^T \mathbb{E}\left[r_t \mathbb{1}\{\mathcal{G}_t\} \mathbb{1}\left\{\sigma(x_t^T \hat{\theta}_{t-1}) \leq \alpha - \beta_{t-1}^{x_t}(\delta_t)\right\}\right]}_{F_{11}} + \underbrace{\sum_{t=1}^T \mathbb{E}\left[r_t \mathbb{1}\{\mathcal{G}_t\} \mathbb{1}\left\{\sigma(x_t^T \hat{\theta}_{t-1}) \geq \alpha + \beta_{t-1}^{x_t}(\delta_t)\right\}\right]}_{F_{12}} \\ &\quad + \underbrace{\sum_{t=1}^T \mathbb{E}\left[r_t \mathbb{1}\{\mathcal{G}_t\} \mathbb{1}\left\{\alpha - \beta_{t-1}^{x_t}(\delta_t) < \sigma(x_t^T \hat{\theta}_{t-1}) < \alpha + \beta_{t-1}^{x_t}(\delta_t)\right\}\right]}_{F_{13}}. \end{aligned}$$

**Case  $F_{11}$ :**  $\hat{y}_t = \sigma(x_t^T \hat{\theta}_{t-1}) \leq \alpha - \beta_{t-1}^{x_t}(\delta_t)$

First of all  $|\hat{y}_t - \alpha| \geq \beta_{t-1}^{x_t}(\delta_t)$  and  $\hat{y}_t \leq \alpha$ , hence,  $a_t = 0$  (Reject) by the behavior of Algorithm 1

This is the case where, estimated distribution  $\hat{y}_t$  is comfortably to the left of flip point, such that, based on Proposition 1, true mean of the distribution  $\sigma(x_t^T \theta^*)$  itself will be on the left side of the flip point.



Since  $\mathcal{G}_t$  happens,

$$\begin{aligned}
 |\sigma(x_t^T \theta^*) - \sigma(x_t^T \hat{\theta}_{t-1})| &\leq \beta_{t-1}^{x_t}(\delta_t) \\
 \sigma(x_t^T \hat{\theta}_{t-1}) - \beta_{t-1}^{x_t}(\delta_t) &\leq \sigma(x_t^T \theta^*) \leq \sigma(x_t^T \hat{\theta}_{t-1}) + \beta_{t-1}^{x_t}(\delta_t) \\
 \sigma(x_t^T \theta^*) &\leq \alpha
 \end{aligned}$$

Hence,  $a_t^* = 0$ .

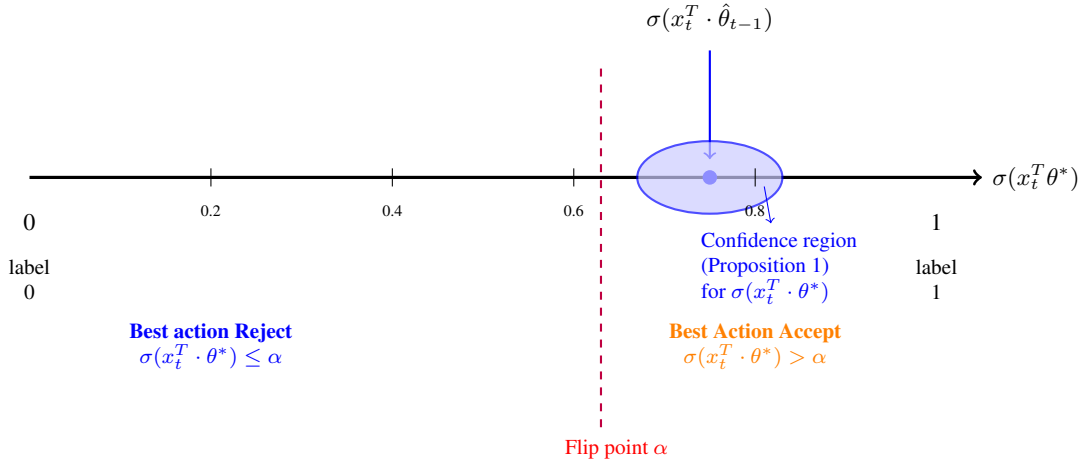
$$a_t = a_t^* = 0.$$

$$F_{11} = 0$$

**Case  $F_{12}$ :**  $\hat{y}_t = \sigma(x_t^T \hat{\theta}_{t-1}) \geq \alpha + \beta_{t-1}^{x_t}(\delta_t)$

First of all  $|\hat{y}_t - \alpha| \geq \beta_{t-1}^{x_t}(\delta_t)$  and  $\hat{y}_t \geq \alpha$ , hence,  $a_t = 1$  (Accept)

This is the case where, estimated distribution  $\hat{y}_t$  is comfortably to the right of flip point, such that, based on Proposition 1, true mean of the distribution  $\sigma(x_t^T \theta^*)$  itself will be on the right side of the flip point.



Since  $\mathcal{G}_t$  happens,

$$\begin{aligned}
|\sigma(x_t^T \theta^*) - \sigma(x_t^T \hat{\theta}_{t-1})| &\leq \beta_{t-1}^{x_t}(\delta_t) \\
\sigma(x_t^T \hat{\theta}_{t-1}) - \beta_{t-1}^{x_t}(\delta_t) &\leq \sigma(x_t^T \theta^*) \leq \sigma(x_t^T \hat{\theta}_{t-1}) + \beta_{t-1}^{x_t}(\delta_t) \\
\sigma(x_t^T \theta^*) &\geq \alpha
\end{aligned}$$

Hence  $a_t^* = 1(\text{Accept})$ .

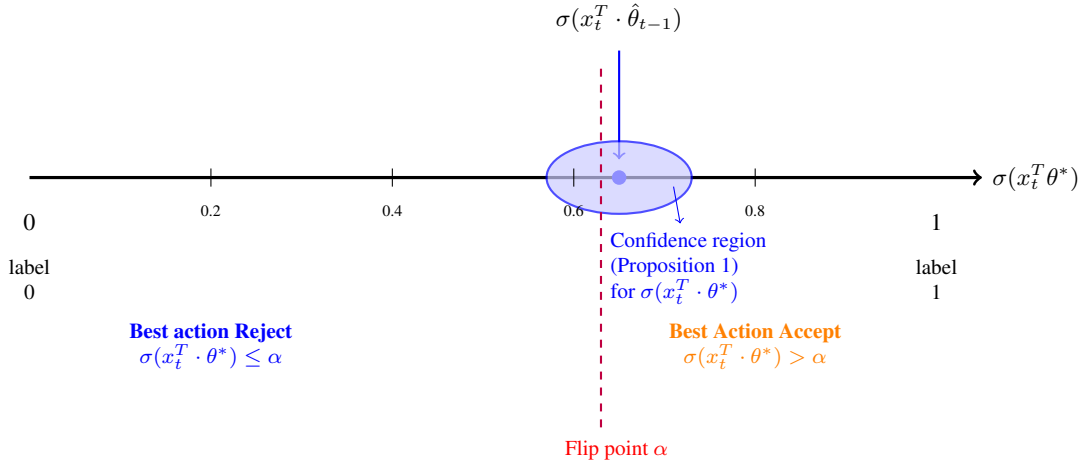
$$a_t = a_t^* = 1.$$

$$F_{12} = 0$$

**Case  $F_{13}$ :**  $\alpha - \beta_{t-1}^{x_t}(\delta_t) < \sigma(x_t^T \hat{\theta}_{t-1}) < \alpha + \beta_{t-1}^{x_t}(\delta_t)$

First of all  $|\hat{y}_t - \alpha| < \beta_{t-1}^{x_t}(\delta_t)$ , hence,  $a_t = 1(\text{Accept})$

This is an uncertain region which requires more exploration. However the nature of the confidence region  $\beta_{t-1}^{x_t}(\delta_t)$  is such that it shrinks with more exploration (because its dependence on  $\|x_t\|_{V_{t-1}^{-1}}^2$  which is a monotonically decreasing function). Hence the regret could be controlled using Elliptical potential lemma.



Since  $\mathcal{G}_t$  happens,

$$\begin{aligned}
|\sigma(x_t^T \theta^*) - \sigma(x_t^T \hat{\theta}_{t-1})| &\leq \beta_{t-1}^{x_t}(\delta_t) \\
\sigma(x_t^T \hat{\theta}_{t-1}) - \beta_{t-1}^{x_t}(\delta_t) &\leq \sigma(x_t^T \theta^*) \leq \sigma(x_t^T \hat{\theta}_{t-1}) + \beta_{t-1}^{x_t}(\delta_t) \\
\alpha - 2\beta_{t-1}^{x_t}(\delta_t) &\leq \sigma(x_t^T \theta^*) \leq \alpha + 2\beta_{t-1}^{x_t}(\delta_t) \\
-2\beta_{t-1}^{x_t}(\delta_t) &\leq \sigma(x_t^T \theta^*) - \alpha \leq 2\beta_{t-1}^{x_t}(\delta_t)
\end{aligned}$$

$$\begin{aligned}
\mathbb{E}_{t-1} [r_t] &= \mathbb{E}_{t-1} [\psi_{a_t}(y_t) - \psi_{a_t^*}(y_t)] \\
&= \mathbb{E}_{t-1} [\psi_1(y_t) - \psi_{a_t^*}(y_t)] \\
&\leq \left| \mathbb{E}_{t-1} [\psi_1(y_t) - \psi_0(y_t)] \right| \\
&= \left| \mathbb{E}_{t-1} [\ell_{10} + y_t(\ell_{11} - \ell_{10}) - \ell_{01}y] \right| \\
&= \left| \ell_{10} + \mathbb{E}_{t-1} [y_t] (\ell_{11} - \ell_{10}) - \ell_{01} \mathbb{E}_{t-1} [y_t] \right| \\
&= \left| \ell_{10} + \sigma(x_t^T \theta^*) (\ell_{11} - \ell_{10}) - \ell_{01} \sigma(x_t^T \theta^*) \right| \\
&= \left| \left( \sigma(x_t^T \theta^*) - \alpha \right) (\ell_{11} - \ell_{10}) - \ell_{01} \left( \sigma(x_t^T \theta^*) - \alpha \right) \right| \quad (\text{definition of } \alpha) \\
&\leq 2\beta_{t-1}^{x_t}(\delta_t) \max(\ell_{10}, \ell_{01} - \ell_{11})
\end{aligned}$$

$$\begin{aligned}
F_{13} &= \sum_{t=1}^T \mathbb{E} \left[ r_t \mathbb{1} \{ \mathcal{G}_t \} \mathbb{1} \left\{ \alpha - \beta_{t-1}^{x_t}(\delta_t) < \sigma(x_t^T \hat{\theta}_{t-1}) < \alpha + \beta_{t-1}^{x_t}(\delta_t) \right\} \right] \\
&\leq \sum_{t=1}^T \mathbb{E} [r_t \mathbb{1} \{a_t = 1\}] \\
&= \sum_{t=1}^T \mathbb{E} [\mathbb{E}_{t-1} [r_t] \mathbb{1} \{a_t = 1\}] \\
&\leq 2 \max(\ell_{10}, \ell_{01} - \ell_{11}) \mathbb{E} \left[ \sum_{t=1}^T \beta_{t-1}^{x_t}(\delta_t) \mathbb{1} \{a_t = 1\} \right] \\
&= 2 \max(\ell_{10}, \ell_{01} - \ell_{11}) \mathbb{E} \left[ \sum_{t=1}^T \frac{2k_\sigma R_{\max}}{c_\sigma} \|x_t\|_{V_{t-1}^{-1}} \sqrt{\left(3 + 2 \log \left(1 + \frac{2}{\lambda}\right)\right) \cdot 2d \log(N_{t-1}) \cdot \log \left(\frac{d}{\delta_t}\right)} (\delta_t) \mathbb{1} \{a_t = 1\} \right] \\
&= 2 \max(\ell_{10}, \ell_{01} - \ell_{11}) \mathbb{E} \left[ \sum_{t=1}^T \frac{2k_\sigma R_{\max}}{c_\sigma} \|x_t\|_{V_{t-1}^{-1}} \sqrt{\left(3 + 2 \log \left(1 + \frac{2}{\lambda}\right)\right) \cdot 2d \log(T) \cdot \log(dT)} (\delta_t) \mathbb{1} \{a_t = 1\} \right] \\
&= \frac{4 \max(\ell_{10}, \ell_{01} - \ell_{11}) k_\sigma R_{\max}}{c_\sigma} \sqrt{\left(3 + 2 \log \left(1 + \frac{2}{\lambda}\right)\right) \cdot 2d \log(T) \cdot \log(Td)} \mathbb{E} \left[ \sum_{t=1}^T \|x_t\|_{V_{t-1}^{-1}} \mathbb{1} \{a_t = 1\} \right] \\
&\leq \frac{4\sqrt{T} \max(\ell_{10}, \ell_{01} - \ell_{11}) k_\sigma R_{\max}}{c_\sigma} \sqrt{\left(3 + 2 \log \left(1 + \frac{2}{\lambda}\right)\right) \cdot 2d \log(T) \cdot \log(Td)} \sqrt{\mathbb{E} \left[ \sum_{t=1}^T \|x_t\|_{V_{t-1}^{-1}}^2 \mathbb{1} \{a_t = 1\} \right]} \\
&\quad \quad \quad (\text{Cauchy-Schwartz})
\end{aligned}$$

By applying Lemma 8,

$$\begin{aligned}
F_{13} &\leq \left( \sqrt{2d \log(1 + \frac{T}{d\lambda})} \right) \frac{4\sqrt{T} \max(\ell_{10}, \ell_{01} - \ell_{11}) k_\sigma R_{\max}}{c_\sigma} \sqrt{\left( 3 + 2 \log \left( 1 + \frac{2}{\lambda} \right) \right) \cdot 2d \log(T) \cdot \log(Td)} \\
&= O \left( \frac{(1 + e^{C_{\max}})^2}{e^{C_{\max}}} d\sqrt{T} \log^{\frac{3}{2}}(T) \right) \quad (\text{Ignoring logarithmic terms in } d)
\end{aligned}$$

Hence,

$$\begin{aligned}
\mathbb{E} [\text{Reg}_T] &\leq \left( \sqrt{2d \log(1 + \frac{T}{d\lambda})} \right) \frac{4\sqrt{T} \max(\ell_{10}, \ell_{01} - \ell_{11}) k_\sigma R_{\max}}{c_\sigma} \sqrt{\left( 3 + 2 \log \left( 1 + \frac{2}{\lambda} \right) \right) \cdot 2d \log(T) \cdot \log(Td)} \\
&\quad + \log T \\
&= O \left( \frac{(1 + e^{C_{\max}})^2}{e^{C_{\max}}} d\sqrt{T} \log^{\frac{3}{2}}(T) \right) \\
&\quad (\text{Ignoring logarithmic terms in } d)
\end{aligned}$$

## C.2 Instance dependent regret analysis

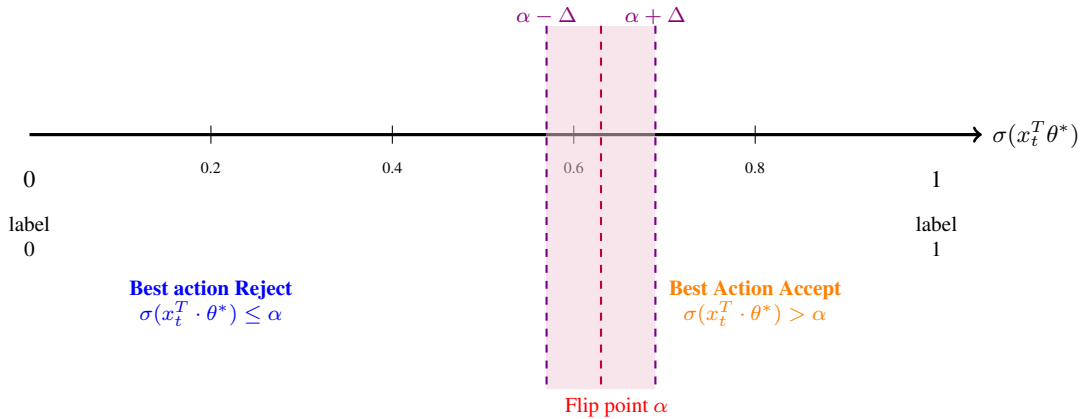
**Theorem 2** (LogCBPSide-AT instance dependent regret bound). *For  $\delta_t = \frac{1}{t}$ , with  $\forall x \in \mathcal{X} \ \|x\|_2 \leq 1$  and  $\sup_{x \in \mathcal{X}} x^T \theta^* \leq C_{\max}$ , and  $0 < \Delta \leq \min_{x \in \mathcal{X}} |\alpha - \sigma(x^T \cdot \theta^*)|$  the expected cumulative regret of the LogCBPSide-AT algorithm is bounded as:*

$$\mathbb{E} [\text{Reg}_T] \leq O\left(\frac{1}{\Delta} d^2 \log^2 T\right)$$

(ignoring doubly logarithmic factors and  $\text{polylog}(\frac{1}{\Delta})$ )

Let define the problem dependent parameter  $\Delta > 0$  as,

$$\Delta \leq \min_{x \in \mathcal{X}} |\alpha - \sigma(x^T \cdot \theta^*)|$$



This defines how close  $y_t^* = \sigma(x_t^T \cdot \theta^*)$  is to the decision boundary  $\alpha$ . The closer, the problem instance become more difficult owing to the fact that we need more exploration to distinguish it. In the figure above, there are no instances of  $x_t$  falls into the shaded region.

With this additional assumption, we can refine our previous analysis.

$F_2, F_{11}, F_{12}$  all go though for this case as well. We skip them and move to the analysis of  $F_{13}$ .

**Case  $F_{13}$ :**  $\alpha - \beta_{t-1}^{x_t}(\delta_t) < \sigma(x_t^T \hat{\theta}_{t-1}) < \alpha + \beta_{t-1}^{x_t}(\delta_t)$

First of all  $|\hat{y}_t - \alpha| < \beta_{t-1}^{x_t}(\delta_t)$ , hence,  $a_t = 1(\text{Accept})$

Since  $\mathcal{G}_t$  happens,

$$\begin{aligned}
|\sigma(x_t^T \theta^*) - \sigma(x_t^T \hat{\theta}_{t-1})| &\leq \beta_{t-1}^{x_t}(\delta_t) \\
\sigma(x_t^T \hat{\theta}_{t-1}) - \beta_{t-1}^{x_t}(\delta_t) &\leq \sigma(x_t^T \theta^*) \leq \sigma(x_t^T \hat{\theta}_{t-1}) + \beta_{t-1}^{x_t}(\delta_t) \\
\alpha - 2\beta_{t-1}^{x_t}(\delta_t) &\leq \sigma(x_t^T \theta^*) \leq \alpha + 2\beta_{t-1}^{x_t}(\delta_t) \\
-2\beta_{t-1}^{x_t}(\delta_t) &\leq \sigma(x_t^T \theta^*) - \alpha \leq 2\beta_{t-1}^{x_t}(\delta_t).
\end{aligned}$$

$$\begin{aligned}
\mathbb{E}_{t-1}[r_t] &= \mathbb{E}_{t-1}[\psi_{a_t}(y_t) - \psi_{a_t^*}(y_t)] \\
&= \mathbb{E}_{t-1}[\psi_1(y_t) - \psi_{a_t^*}(y_t)] \\
&\leq \left| \mathbb{E}_{t-1}[\psi_1(y_t) - \psi_0(y_t)] \right| \\
&= \left| \mathbb{E}_{t-1}[\ell_{10} + y_t(\ell_{11} - \ell_{10}) - \ell_{01}y] \right| \\
&= \left| [\ell_{10} + \mathbb{E}_{t-1}[y_t](\ell_{11} - \ell_{10}) - \ell_{01}\mathbb{E}_{t-1}[y_t]] \right| \\
&= \left| [\ell_{10} + \sigma(x_t^T \theta^*)(\ell_{11} - \ell_{10}) - \ell_{01}\sigma(x_t^T \theta^*)] \right| \\
&= \left| \left[ (\sigma(x_t^T \theta^*) - \alpha)(\ell_{11} - \ell_{10}) - \ell_{01}(\sigma(x_t^T \theta^*) - \alpha) \right] \right| \quad (\text{definition of } \alpha) \\
&\leq 2\beta_{t-1}^{x_t}(\delta_t) \max(\ell_{10}, \ell_{01} - \ell_{11})
\end{aligned}$$

$$\begin{aligned}
F_{13} &= \sum_{t=1}^T \mathbb{E} \left[ r_t \mathbb{1} \{ \mathcal{G}_t \} \mathbb{1} \left\{ \alpha - \beta_{t-1}^{x_t}(\delta_t) < \sigma(x_t^T \hat{\theta}_{t-1}) < \alpha + \beta_{t-1}^{x_t}(\delta_t) \right\} \right] \\
&\leq \sum_{t=1}^T \mathbb{E} \left[ r_t \mathbb{1} \{ a_t = 1 \} \mathbb{1} \left\{ -2\beta_{t-1}^{x_t}(\delta_t) \leq \sigma(x_t^T \theta^*) - \alpha \leq 2\beta_{t-1}^{x_t}(\delta_t) \right\} \right] \\
&= \sum_{t=1}^T \mathbb{E} \left[ r_t \mathbb{1} \{ a_t = 1 \} \mathbb{1} \left\{ |\sigma(x_t^T \theta^*) - \alpha| \leq 2\beta_{t-1}^{x_t}(\delta_t) \right\} \right] \\
&= \sum_{t=1}^T \mathbb{E} \left[ r_t \mathbb{1} \{ a_t = 1 \} \mathbb{1} \left\{ |\sigma(x_t^T \theta^*) - \alpha| \leq 2\beta_{t-1}^{x_t}(\delta_t) \right\} \mathbb{1} \left\{ \beta_{t-1}^{x_t}(\delta_t) < \frac{\Delta}{2} \right\} \right] \\
&\quad + \sum_{t=1}^T \mathbb{E} \left[ r_t \mathbb{1} \{ a_t = 1 \} \mathbb{1} \left\{ |\sigma(x_t^T \theta^*) - \alpha| \leq 2\beta_{t-1}^{x_t}(\delta_t) \right\} \mathbb{1} \left\{ \beta_{t-1}^{x_t}(\delta_t) \geq \frac{\Delta}{2} \right\} \right] \\
&= \sum_{t=1}^T \mathbb{E} \left[ r_t \mathbb{1} \{ a_t = 1 \} \mathbb{1} \left\{ |\sigma(x_t^T \theta^*) - \alpha| < \Delta \right\} \right] \\
&\quad + \sum_{t=1}^T \mathbb{E} \left[ r_t \mathbb{1} \{ a_t = 1 \} \mathbb{1} \left\{ |\sigma(x_t^T \theta^*) - \alpha| \leq 2\beta_{t-1}^{x_t}(\delta_t) \right\} \mathbb{1} \left\{ \beta_{t-1}^{x_t}(\delta_t) \geq \frac{\Delta}{2} \right\} \right] \\
&= 0 \quad \text{(By definition of } \Delta \text{)} \\
&\quad + \sum_{t=1}^T \mathbb{E} \left[ r_t \mathbb{1} \{ a_t = 1 \} \mathbb{1} \left\{ |\sigma(x_t^T \theta^*) - \alpha| \leq 2\beta_{t-1}^{x_t}(\delta_t) \right\} \mathbb{1} \left\{ \beta_{t-1}^{x_t}(\delta_t) \geq \frac{\Delta}{2} \right\} \right] \\
&= \sum_{t=1}^T \mathbb{E} \left[ r_t \mathbb{1} \{ a_t = 1 \} \mathbb{1} \left\{ |\sigma(x_t^T \theta^*) - \alpha| \leq 2\beta_{t-1}^{x_t}(\delta_t) \right\} \mathbb{1} \left\{ \beta_{t-1}^{x_t}(\delta_t) \geq \frac{\Delta}{2} \right\} \right] \\
&\leq 2 \max(\ell_{10}, \ell_{01} - \ell_{11}) \sum_{t=1}^T \mathbb{E} \left[ \mathbb{1} \{ a_t = 1 \} \beta_{t-1}^{x_t}(\delta_t) \mathbb{1} \left\{ \beta_{t-1}^{x_t}(\delta_t) \geq \frac{\Delta}{2} \right\} \right]
\end{aligned}$$

From Proposition 1,

$$\begin{aligned}
\beta_{t-1}^{x_t}(\delta_t) &= \frac{2k_\sigma R_{\max}}{c_\sigma} \|x_t\|_{V_{t-1}^{-1}} \sqrt{\left(3 + 2 \log \left(1 + \frac{2}{\lambda}\right)\right) \cdot 2d \log(N_{t-1}) \cdot \log\left(\frac{d}{\delta_t}\right)} \\
&\leq \frac{2k_\sigma R_{\max}}{c_\sigma} \|x_t\|_{V_{t-1}^{-1}} \sqrt{\left(3 + 2 \log \left(1 + \frac{2}{\lambda}\right)\right) \cdot 2d \log T \cdot \log(dT)} \\
&:= h(T) \cdot \|x_t\|_{V_{t-1}^{-1}} \\
\beta_{t-1}^{x_t}(\delta_t) \geq \frac{\Delta}{2} &\implies \|x_t\|_{V_{t-1}^{-1}} \geq \frac{\Delta}{2h(T)} \\
&\quad \text{(Let } h(T) := \frac{2k_\sigma R_{\max}}{c_\sigma} \sqrt{\left(3 + 2 \log \left(1 + \frac{2}{\lambda}\right)\right) \cdot 2d \log T \cdot \log(dT)} \text{)}
\end{aligned}$$

$$\begin{aligned}
F_{13} &\leq 2 \max(\ell_{10}, \ell_{01} - \ell_{11}) \sum_{t=1}^T \mathbb{E} \left[ \mathbb{1} \{a_t = 1\} \beta_{t-1}^{x_t}(\delta_t) \mathbb{1} \left\{ \|x_t\|_{V_{t-1}^{-1}} \geq \frac{\Delta}{2h(T)} \right\} \right] \\
&\leq 2 \max(\ell_{10}, \ell_{01} - \ell_{11}) h(T) \sum_{t=1}^T \mathbb{E} \left[ \mathbb{1} \{a_t = 1\} \|x_t\|_{V_{t-1}^{-1}} \mathbb{1} \left\{ \|x_t\|_{V_{t-1}^{-1}} \geq \frac{\Delta}{2h(T)} \right\} \right]
\end{aligned}$$

By applying peeling method,

$$\begin{aligned}
F_{13} &\leq 2 \max(\ell_{10}, \ell_{01} - \ell_{11}) h(T) \sum_{t=1}^T \mathbb{E} \left[ \mathbb{1} \{a_t = 1\} \|x_t\|_{V_{t-1}^{-1}} \mathbb{1} \left\{ \|x_t\|_{V_{t-1}^{-1}} \geq \frac{\Delta}{2h(T)} \right\} \right] \\
&= 2 \max(\ell_{10}, \ell_{01} - \ell_{11}) h(T) \sum_{t=1}^T \sum_{\ell=1}^{\infty} \mathbb{E} \left[ \mathbb{1} \{a_t = 1\} \|x_t\|_{V_{t-1}^{-1}} \mathbb{1} \left\{ 2^{\ell+1} \frac{\Delta}{2h(T)} \geq \|x_t\|_{V_{t-1}^{-1}} \geq \frac{\Delta}{2h(T)} \cdot 2^{\ell} \right\} \right] \\
&\leq 2 \max(\ell_{10}, \ell_{01} - \ell_{11}) h(T) \sum_{t=1}^T \sum_{\ell=1}^{\infty} \mathbb{E} \left[ \mathbb{1} \{a_t = 1\} 2^{\ell+1} \frac{\Delta}{2h(T)} \mathbb{1} \left\{ 2^{\ell+1} \frac{\Delta}{2h(T)} \geq \|x_t\|_{V_{t-1}^{-1}} \geq \frac{\Delta}{2h(T)} \cdot 2^{\ell} \right\} \right] \\
&\leq \max(\ell_{10}, \ell_{01} - \ell_{11}) \sum_{t=1}^T \sum_{\ell=1}^{\infty} \mathbb{E} \left[ 2^{\ell+1} \Delta \mathbb{1} \left\{ 2^{\ell+1} \frac{\Delta}{2h(T)} \geq \|x_t\|_{V_{t-1}^{-1}} \geq \frac{\Delta}{2h(T)} \cdot 2^{\ell} \right\} \right] \\
&= \max(\ell_{10}, \ell_{01} - \ell_{11}) \sum_{\ell=1}^{\infty} 2^{\ell+1} \Delta \sum_{t=1}^T \mathbb{E} \left[ \mathbb{1} \left\{ 2^{\ell+1} \frac{\Delta}{2h(T)} \geq \|x_t\|_{V_{t-1}^{-1}} \geq \frac{\Delta}{2h(T)} \cdot 2^{\ell} \right\} \right] \\
&\leq \max(\ell_{10}, \ell_{01} - \ell_{11}) \sum_{\ell=1}^{\infty} 2^{\ell+1} \Delta \sum_{t=1}^T \mathbb{E} \left[ \mathbb{1} \left\{ \|x_t\|_{V_{t-1}^{-1}} \geq \frac{\Delta}{2h(T)} \cdot 2^{\ell} \right\} \right] \\
&= \max(\ell_{10}, \ell_{01} - \ell_{11}) \sum_{\ell=1}^{\infty} 2^{\ell+1} \Delta \mathbb{E} \left[ \sum_{t=1}^T \mathbb{1} \left\{ \|x_t\|_{V_{t-1}^{-1}}^2 \geq \frac{\Delta^2}{4h^2(T)} \cdot 2^{2\ell} \right\} \right] \\
&\leq \max(\ell_{10}, \ell_{01} - \ell_{11}) \sum_{\ell=1}^{\infty} 2^{\ell+1} \Delta \mathbb{E} \left[ \frac{12dh^2(T)}{2^{2\ell}\Delta^2} \log \left( 1 + \frac{8h^2(T)}{\lambda 2^{2\ell}\Delta^2} \right) \right] \quad (\text{Lemma 9 (EPC)}) \\
&\leq \frac{1}{\Delta} 24dh^2(T) \max(\ell_{10}, \ell_{01} - \ell_{11}) d \log \left( 1 + \frac{8h^2(T)}{\lambda 4\Delta^2} \right) \sum_{\ell=1}^{\infty} 2^{-\ell} \\
&\leq \frac{1}{\Delta} 24dh^2(T) \max(\ell_{10}, \ell_{01} - \ell_{11}) d \log \left( 1 + \frac{8h^2(T)}{\lambda 4\Delta^2} \right) \\
&\quad (\text{here } h(T) = \frac{2k_{\sigma} R_{\max}}{c_{\sigma}} \sqrt{\left( 3 + 2 \log \left( 1 + \frac{2}{\lambda} \right) \right) \cdot 2d \log T \cdot \log(dT)})
\end{aligned}$$

Hence,

$$\begin{aligned}
\mathbb{E} [Reg_T] &\leq \frac{1}{\Delta} 24dh^2(T) \max(\ell_{10}, \ell_{01} - \ell_{11}) d \log \left( 1 + \frac{8h^2(T)}{\lambda 4\Delta^2} \right) + O(\log(T)) \\
&\quad (\text{here } h(T) = \frac{2k_{\sigma} R_{\max}}{c_{\sigma}} \sqrt{\left( 3 + 2 \log \left( 1 + \frac{2}{\lambda} \right) \right) \cdot 2d \log T \cdot \log(dT)}) \\
&= O\left(\frac{1}{\Delta} d^2 \log^2 T\right) \quad (\text{ignoring doubly logarithmic factors and polylog}(\frac{1}{\Delta}))
\end{aligned}$$



## D Experiment Details

We call our framework AppleTeA: Test-Time Adaptive LLM Cascading via Logistic Apple Tasting, which is presented in Algorithm D

---

### Algorithm 4 AppleTeA

---

```

Input:  $\mathcal{M}_s, \mathcal{M}_\ell, n, \phi, \alpha, \gamma, \lambda, \mu$ , algorithm Alg
if Alg = "LogCB-AT" then
    Instantiate  $\mathcal{A} = \text{LogCB-AT}(\gamma, \mu)$ 
else
    Instantiate  $\mathcal{A} = \text{LogCBPSide-AT}(\lambda, \alpha)$ 
end if
for  $t = 1, \dots, n$  do
    Observe the query  $q_t$ 
    Run SLM  $r_t = \mathcal{M}_s(q_t)$ 
    Extract the answer:  $\hat{y}_{w,t} = \text{answer}(r_t)$ 
    Compute the context:  $x_t = \phi(q_t, r_t)$ 
    Run single step of algorithm  $\mathcal{A}$  and receive the action recommendation  $a_t$ 
    if  $a_t = 0$  (Reject) then
        output  $\hat{y}_{w,t}$ 
    else
        Run the strong LLM:  $r_t = \mathcal{M}_\ell(q_t)$ 
        Extract the answer:  $\hat{y}_{s,t} = \text{answer}(r_t)$ 
        Compute the context:  $x_t = \phi(q_t, r_t)$ 
        Update  $\mathcal{A}$  with context  $x_t$  and reward  $= \mathbb{1}\{\hat{y}_{s,t} \neq \hat{y}_{w,t}\}$ 
        output  $\hat{y}_{s,t}$ 
    end if
end for

```

---

We have established that both LogCB-AT and LogCBPSide-AT attain sub-linear regret with respect to the optimal policy. This regret bound directly translates into a cumulative cost guarantee for AppleTeA. By carefully constructing the loss matrix to encode two competing cost components 1) the invocation cost of the strong LLM (i.e., token-level inference cost) and 2) the price of error incurred when the weak SLM is invoked in contexts where prediction failure is likely; AppleTeA is equipped to navigate this cost-quality tradeoff in a principled manner. Consequently, AppleTeA achieves optimal performance in expectation, a guarantee that follows directly from the formulation of the model-selection problem as an Apple Tasting problem and the sub-linear regret guarantees of the underlying core algorithms.

**Context Construction.** To enable effective decision-making, we construct the context vector  $x_t \in \mathbb{R}^d$  by capturing the internal state and the uncertainty score of the SLM model. Specifically,  $x_t$  is formed by concatenating the dimensionally-reduced hidden state vector of the final token from  $\mathcal{M}_s$  and its generative uncertainty score. The uncertainty score can be quantified using either the maximum probability of the generated tokens or the logit margin between the top two candidate answers. This context provides a rich signal for the the proposed methods to estimate the probability of the LLM model altering the final answer.

**Datasets and Models.** We adopt Pythia-2.8B [Biderman et al., 2023] and LLaMA2-13B [Touvron et al., 2023] as  $\mathcal{M}_s$  and  $\mathcal{M}_\ell$ , respectively. We evaluate the effectiveness of our algorithms on the commonsense reasoning benchmark: ARC-Easy [Clark et al., 2018], OpenBookQA [Mihaylov et al., 2018] and BoolQ [Sakaguchi et al., 2021]

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Justification: LLMs were not used as a core component of the methodology in this research.

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